Joint SDG Fund - 1st Call on SDG Financing

Component 2 Joint Programme Document

A. COVER PAGE

1. Fund Name: Joint SDG Fund

2. MPTFO Project Reference Number:

3. Joint programme title: Green Finance Facility to Improve Air Quality and Combat Climate Change in North Macedonia

4. Short title: North Macedonia Green Finance Facility

5. Country and region: Republic of North Macedonia

6. Resident Coordinator: Ms. Rossana Dudziak, rossana.dudziak@un.org

7. RCO Joint Programme focal point: Ms. Lilian Kandikjan, Partnership and Development Finance, lilian.kandikjan@un.org

8. Lead agency Joint Programme focal point: UNDP North Macedonia, Ms. Sanja Bojanic, Deputy Resident Representative, sanja.bojanic@undp.org

9. Government Joint Programme focal point Cabinet of the Deputy Prime Minister of Economic Affairs, Mr. Fatmir Bytyqi, Deputy Prime Minister of Economic Affairs

10. Type of financial intervention: Loans with Performance-Based Payments

11. Short description:
With the support from the Joint SDG Fund, this project will set-up and operationalize a Green Financing Facility (GFF), a new financial vehicle, that will provide access to affordable green financing for SMEs and marketable but underserved target groups of individuals/households for investments in renewable energy (RE) and energy efficiency (EE) solutions. The GFF enabled accelerated adoption of RE and EE solutions will contribute to the reduction of greenhouse gas emissions and air pollution and strengthen the local ecosystem for RE & EE green finance.

12. Keywords: green finance, renewable energy financing, energy efficiency financing, air pollution reduction, climate change mitigation, GHG emissions reduction, remittances.

13. Overview of budget

Joint SDG Fund contribution[1]	**$7,800,000**[2]
Co-funding EUR 27M EBRD	$31,400,000[3]
Co- funding by local banks	$6,500,000
Co-funding from the Government of North Macedonia	$ 800,000
TOTAL	**$46,500,000**

14. Timeframe:

Start date	**End date**	**Duration** (in months)
July 2022	July 2026	48 months

15. Gender Equality Marker: GEM 2.3

16. Participating UN Organizations (PUNO) and Partners:

16.1 PUNOs

Convening agency:

- **UNDP:** Mr. Armen Grigoryan, Resident Representative, armen.grigoryan@undp.org, +389-2-324-9500

Other PUNOs:

- **IOM**: Ms. Sonja Bozinovska Petrusevska, Head of Office, sbozinovska@iom.int, +389-2-3095-477
- **UNECE**: Mr. Alan Esser, Officer-in-Charge, Executive Office, Programme Management & Support Services Division (PMSSD), essera@un.org, +41 (0) 22 917 79 05

16.2 Partners

- National authorities:
  - **Cabinet of the Deputy Prime Minister of Economic Affairs**, Mr. Fatmir Bytyqi, Deputy Prime Minister of Economic Affairs, fatmir.bytyqi@gs.gov.mk;+389-2-322-0095
- International Financial Institutions:
  - **EBRD North Macedonia Office**, Mr. Andi Aranitasi, Head of EBRD North Macedonia, AranitA@ebrd.com, +389-2-329-7800
- Private sector:
  - Local Banks, partners for Outcome 1: **Komercijalna Banka, NLB Bank Skopje, Sparkasse Bank Macedonia, Stopanska Banka Skopje, and Procredit Bank Macedonia**
  - Local Banks, partners for Outcome 2: **Silk Road Bank, Stopanska Banka Bitola and Halk Bank**
- Other partners:
  - **Ministry of Economy**
  - **Ministry of Environment and Physical Planning**
  - **Ministry of Finance**

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[1] *All figures in the table have been rounded.*

[2] *Government of North Macedonia provided co-financing in the amount of $0.8M for the performance based payments (PBPs) and EBRD matched that with additional resources. However, additional amount was needed for TA and the Joint SDG Fund contribution was increased for that amount. TA is calculated at the level of 5% but it will be subject to contracting and if the contracted amount is less, the money will be channeled to PBPs. Also, the amount requested from the Joint SDG Fund can be reduced to the original request, but it would mean that money which is already committed by EBRD will have to be forgone.*

[3] *US dollar amount calculated using the UN Exchange Rate at 01.Oct 2021 of 1 Euro = USD 1.16*

UN SIGNATURE PAGE

Resident Coordinator *Rossana Dudziak* *Date* *Signature*
Participating UN Organization (lead/convening) *UNDP* *Armen Grigoryan, Resident Representative* *Date* *Signature and seal*
Participating UN Organization *IOM* *Sonja Bozinovska Petrushevska, Head of Office* *Date* *Signature and seal*
Participating UN Organization *UNECE* *Alan Esser, Officer-in-Charge, Executive Office* *Date* *Signature and seal*

GOVERNMENT SIGNATURE PAGE

National Coordinating Authority
Cabinet of the Deputy Prime Minister for Economic Affairs
Fatmir Bytyqi, Deputy Prime Minister for Economic Affairs

Date
Signature and seal

☒ **Official support letter attached**

B. STRATEGIC FRAMEWORK

1. Call for Joint Programmes: SDG Financing (2020) – Component 2

2. Programme Outcome [pre-selected]

- Additional financing leveraged to accelerate SDG achievement (Joint SDG Fund Outcome 2)

3. UNSDCF Outcomes and Outputs

3.1 The joint programme will contribute primarily to **UNSDCF Outcome 3**: By 2025, people in North Macedonia benefit from ambitious climate action, sustainably managed natural resources and well-preserved biodiversity through good environmental governance and disaster resilient communities.

3.2 **UNSDCF Output 3.1:** Enabling policy and regulatory environment strengthened for low-emission development and resilience to climate change.

4. SDG Targets directly addressed by the Joint Programme

4.1 List of goals and targets

SDG Goal 7: Affordable and clean energy

Target 7.2: By 2030, increase substantially the share of RE in the global energy mix.
Target 7.3: By 2030, double the global rate of improvement in energy efficiency.
Target 7.a: By 2030, enhance international cooperation to facilitate access to clean energy research and technology, including renewable energy, energy efficiency and advanced and cleaner fossil-fuel technology, and promote investment in energy infrastructure and clean energy technology.

SDG Goal 13: Take urgent action to combat climate change and its impacts.

Target 13.2.2.: Total greenhouse gas emissions per year

4.2 Expected SDG impact: This project is aligned to and supports the achievement of SDG and the UNSDCF targets. It will result in 70,068 MWh of annual energy savings, 80,510 tons of CO2-eq emissions avoided, and 10.7 MW in new renewable energy capacity. Furthermore, enabling access to affordable financing for renewable energy and energy efficiency solutions will contribute to advancements towards achieving SDGs 7 and 13. The project will directly contribute to SDG 7 (Affordable and Clean Energy) by making the acquisition of clean energy more affordable and by lowering energy consumption. The project will directly contribute to SDG 13 as it will significantly reduce GHG emissions. More broadly, as emissions are closely linked to resource use, this project supports 'green growth' as the future development pathway that promotes resource use efficiency and low-carbon development.

5. Relevant objectives from the national SDG framework

North Macedonia has the following relevant objectives that this project addresses:

SDG 7 Affordable and Clean Energy: Reduce primary energy consumption by 52% and final energy consumption by 28% by 2040. Reduce GHG emissions by 62% by 2040 in comparison with 2005 levels while strongly increasing renewable energy sources. Streamline energy transition technologies into national research and innovation priorities.

SDG 13 Climate Action: North Macedonia submitted its enhanced Nationally Determined Contributions (NDCs) in April 2021 so as to be in-line with the objectives of the newly adopted National Strategy for Energy Development 2040. The new enhanced NDCs foresee an ambitious target of 51% greenhouse gas emissions reduction by 2030, compared to base scenario.

The preparation of the Law and Long-Term Strategy on Climate Action is ongoing while the long-term strategy on climate change is adopted by the Government in September 2021. The country has drafted a National Energy and Climate Plan in-line with the five dimensions of the Energy Community which is expected to be approved by the end of 2021.

The National Energy and Climate Plan covers the period from 2021 to 2030 and proposes 63 policies and measures for achieving the set targets and objectives for each of the five pillars of the National Strategy for Energy Development 2040. In terms of energy efficiency, for instance, North Macedonia is planning to implement several policies and measures over the period 2020-2040 to reduce energy consumption in buildings (households, commercial and public), industry and transport, and to reduce the losses in the energy transformation, transmission and distribution network.

6. Trans-boundary and/or regional issues

The main electricity power plant in North Macedonia is the coal-based Thermal Power Plant in Bitola, which is located about 20 kilometers from the border with Greece. As such, energy production results in pollution that spills/carries over into Greece. By reducing the energy demand in North Macedonia through energy efficiency and utilization of renewable sources, gradual decommissioning of the coal power plant will be supported, thereby reducing transboundary air pollution transmission. Moreover, North Macedonia has a high dependence on electricity imports at around 30 percent of total consumption4. Accordingly, increasing the RE contribution to the nationally produced energy mix will help with diversification of the domestic energy supply thereby reducing dependence on energy imports, especially when combined with the energy efficiency measures.

[4] *https://bankwatch.org/beyond-coal/the-energy-sector-in-macedonia#:~:text=North%20Macedonia%20relies%20predominantly%20on,the%20total%20domestic%20electricity%20demand.*

C. JOINT PROGRAMME DESCRIPTION

1. Baseline and Situation Analysis

1.1 Problem statement

In North Macedonia, more than 60% of electricity generation is coal-fired[5]. Furthermore, only a small percentage of the population are utilizing renewable energy and energy-efficiency solutions. Combined, these factors contribute to significant GHG emissions (7.5 million tons from the burning of fossil fuels[6]) and high levels of air pollution.

The World Health Organization recently ranked the capital city of Skopje as Europe's third most polluted city. In 2018, Skopje's average annual air pollution stood at 40 µg/m3, which is 60% higher than EU guidelines and four times higher than WHO guidelines. North Macedonia air pollution has especially high levels of particulate matter (PM). In fact, the country has the highest concentration (51 µg/m37) of PM2.5 among the 41 European countries[8].

Climate change not only exacerbates air pollution problems, it also increases existing gender inequalities as it leads to increases in the energy and economic poverty of women who have higher rates of unemployment and frequently are the head of single-parent households with several dependents.[9]

North Macedonia has significant renewable energy potential, especially solar energy. The estimated total national technical potential for installed solar energy is 14,000 MW[10],however, the current installed solar generation is only 30 MW[11] or around 2% of the national technical potential. Until recently the legal framework was not supportive of small producers of solar energy or for producer-consumer schemes. However, the new Law on Energy and the Rulebook on Renewable Energy Sources has changed this dynamic by providing a supported enabling environment for the utilization of solar energy which is yet to be capitalized on.

North Macedonia's industrial sector is a key economic driver. However, its energy intensity is nearly 4 times higher than the EU28 average. High energy intensity is a function of low levels of investment in energy efficiency and energy management. This high energy intensity negatively affects the competitiveness of Macedonian products on the European market.
Although electricity prices are constantly increasing and new energy law has fully liberalized the electricity market, the national electricity price is still lower than the EU average. Considering North Macedonia's accession to the EU process and liberalization of the country's energy market, electricity prices are expected to continue to increase, which put more financial pressure on households and the business sector.

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[5] *SDG Voluntary National Review July 2020*
[6] *National GHG inventory, December 2020*
[7] *Data for 2016-2017 North Macedonia*
[8] *According to the Air quality in Europe — 2019 Report as measured at urban and suburban settings*
[9] *Women have higher rates of unemployment and frequently are the head of single-parent households with several dependents.*
[10] *IRENA - COST-COMPETITIVE RENEWABLE POWER GENERATION: Potential across South East Europe*
[11] *Energy and water services regulatory commission of the Republic of North Macedonia- Annual report 2020*

According the latest Energy Balance (2019) published by the State Statistical Office, the household sector accounts for nearly 24% of national energy consumption. 32.3[12]% of households have any insulation in their homes with 25% of the population unable to keep their homes adequately warm. In addition, 50%[13] of the households use biomass for heating purposes, which makes the household sector the most significant source of PM emissions in the country.

Transitioning to a more sustainable energy mix through increased production of renewable energy (particularly solar) and energy efficient retrofitting of buildings and homes is a major strategic objective of the Government. Efforts have been constrained by low awareness of the options available and the value of RE and EE solutions by SMEs and individuals/households, especially among the groups targeted by this project. Also, SMEs and individuals have also been reluctant to invest in these types of products due to the high upfront cost and the relatively long payback period (in the form of lower utility bills).

Accelerating the country's energy transition requires increasing awareness and a banking sector that provides affordable finance to businesses and households for these types of investments. Current available financial products are limited in scope and access.

1.2 Related interventions

Access to finance for the acquisition of EE and especially RE solutions is still limited in North Macedonia and according to the recently completed study by IRENA[14], availability of affordable finance is among the top barriers for adoption of RE solutions in North Macedonia. The current initiatives for increased access to finance for RE and EE are few, sporadic and focused on public sector EE (World Bank), large scale RE investments financing and financing for EE for the more affluent households (EBRD), while even more limited financing is available for the underserved and vulnerable households for energy efficiency (Habitat for Humanity and Government programme) and financial support for increasing competitiveness for SMEs (Government subsidy programme). In this landscape, SMEs and less affluent households remain largely underserved and unincentivized to invest in clean energy, especially in RE. A more detailed description of related interventions is provided in Annex 1.

1.3 SDGs and targets

The project will contribute to the achievement of the following SDG goals and targets:

SDG Goal 7: Affordable and clean energy

Target 7.2: By 2030, increase substantially the share of RE in the global energy mix.
Target 7.3: By 2030, double the global rate of improvement in energy efficiency.
Target 7.a: By 2030, enhance international cooperation to facilitate access to clean energy research and technology, including renewable energy, energy efficiency and advanced and cleaner fossil-fuel technology, and promote investment in energy infrastructure and clean energy technology.

[12] *State statistical office, Energy consumption in households, 2019*
13 State statistical office, Energy consumption in households, 2019
[14] *IRENA and UNDP (2021), De-risking investments in the Republic of North Macedonia: Renewable energy finance and policy landscape focusing on power, heating and cooling in line with the Macedonian Nationally Determined Contributions on Climate Change, International Renewable Energy Agency, Abu Dhabi.*

SDG Goal 13: Take urgent action to combat climate change and its impacts.

Target 13.2: Integrate climate change measures into national policies, strategies and planning.

Additionally, the project will indirectly drive advances on the following SDGs:

- SDG 3 (Good Health and Well-being): contaminated air was linked to 7.3% of deaths in 2018. Such a major health crisis requires immediate solutions by both public and private actors to resolve the issue
- SDG 8 (Decent Work and Economic Growth): poor health has a considerable impact on economic growth as it increases medical expenditures, decreases work productivity and contributes to a rise in premature deaths. The World Bank estimates that the productivity impact of air pollution is around 0.13% of GDP
- SDG 10 (Reduced Inequalities): the project will pay particular attention to certain groups including female-headed households, single parents, remittance-recipients, returning migrants, Roma and persons with disabilities
- SDG 11 (Sustainable Cities and Communities): the project will target improvements in air quality, especially in the capital city

Baseline data on the SDG targets as well as brief analysis of interlinkages amongst the SDGs and opportunities for systemic change are provided in Section 2.6 Results below.

2. Programme Strategy

2.1 Theory of Change

Challenge

North Macedonia has high levels of air pollution that stem in large part from electricity generation and household heating practices. Electricity generation is around 60% coal-fired, while buildings and homes are highly energy inefficient with limited utilization of RE solutions.

Increases in electricity generation from renewable sources and, utilization of energy efficiency solutions are required. Currently, the utilization of RE and EE solutions is limited as the upfront costs for or of these solutions are considered to be too high for the target consumers. Additionally, there is lack of understanding of the available technology options by the target groups. As such, there is a market gap that requires incentivized financing, and technical assistance to design optimal configurations.

Preferred solution

For North Macedonia to accelerate the widespread adoption of renewable energy and energy efficiency solutions the provision of affordable finance complemented with incentives and technical assistance is required.

By developing and operationalizing a *Green Financing Facility* access to finance for SMEs and households (from the target groups) to purchase RE and EE investments will be significantly enhanced. Doing so will enable the accelerated adoption of these solutions and further lead to a decreased reliance on non-renewable (coal-based) energy production, better energy efficiency solutions and practices in homes and businesses, and reduced energy consumption. The net effect is a reduction in greenhouse gas emissions and contributions to air pollution.

Implementing this solution will create a new paradigm in North Macedonia, and eventually across the Western Balkans, in which the whole ecosystem of banks, product vendors, and

service providers is stimulated to work together with various stakeholders to ensure required financial and technical support that accelerates the transition to RE and EE solutions and enhances local innovation.

This project is transformational in that the Green Financing Facility, as a flexible and lean financial mechanism, connects multiple interventions to unlock private capital flows from financial institutions, SMEs and individuals for the acquisition and deployment of RE and EE technologies that result in reductions in greenhouse gas emissions and air pollution, increases in efficiencies in energy consumption and increase in application of renewable energy solutions, especially solar, in both households and businesses.

Barriers to the solution
The following barriers to the preferred solution have been identified as summarized below:

- ***Barrier 1****:* Local financial institutions have limited interest in providing lending products for the acquisition of RE and EE solutions. This is due to a combination of factors including what is seen as a limited consumer demand, a lack of knowledge about the risk and return scenarios, and limited knowledge about the technical/engineering options and details of designing and installing RE/EE solutions by both banks and borrowers[15].
- ***Barrier 2****:* The willingness of SMEs and individuals to invest in RE/EE solutions without provision of incentives is limited due to the perceived costs as well as low level of awareness about the economic and health benefits of RE and EE solutions.
- ***Barrier 3:*** A low level of absorption of financial instruments by women and women-owned companies. This is due to multiple factors, among which are the low number of women-owned businesses (29%), and their small size (99% micro) and persisting traditional gender roles. Broadly, only 25% of all individual loan recipients are women. The employment rate of women is 39% which is significantly lower than that of men at 60%.
- ***Barrier 4:*** Target groups perceive RE solutions as complicated and costly, viewing them as a cost not an investment. Furthermore, they are not sufficiently familiar with the options for RE solutions.
- ***Barrier 5:*** While ~38% of the remittance recipients[16] utilize those remittances for purchasing EE solutions - spending on RE solutions is constrained by high upfront costs, limited access to financing and a lack of customized advisory support services.

Project approach to overcoming the barriers
The proposed project will contribute to overcoming the barriers including enabling and triggering the unlocking of private capital for investments in renewable energy and energy efficiency solutions.

Barrier 1 will be addressed by a combination of funding from the EBRD in form of credit lines channeled through the local banks to the final users, which will be supported by performance-based grants and TA. The TA will consist of two sets of independent consultants – Project Consultants and Verification Consultants as explained below.

Barrier 2 will be addressed by building awareness of the project and the value of RE and EE solutions through organized workshops and marketing campaigns with partner banks and

[15] *IRENA and UNDP (2021), De-risking investments in the Republic of North Macedonia: Renewable energy finance and policy landscape focusing on power, heating and cooling in line with the Macedonian Nationally Determined Contributions on Climate Change, International Renewable Energy Agency, Abu Dhabi.*
16 Data from respondents of the Behavioral Study

relevant CSOs[17] which will build the confidence of these groups to design and implement solutions.

Barrier 3 will be addressed by focusing on promoting and supporting women to participate in the offerings. Special efforts will be made to target women via women-focused groups.

Barrier 4 will be addressed by providing performance-based payments to the marketable but underserved groups that the project targets: female headed households, single parents, households with persons with disabilities (PWD), Roma households, and employees who were affected by COVID–19.

Barrier 5 will be addressed by providing $0.45M of performance-based payments and $0.15M for technical assistance to remittance recipients and returning migrants. Doing so is expected to incentivise these groups to take a subsidized loan. Furthermore, project consultants will provide technical support to both the banks and borrowers, and marketing efforts will be made to promote the value of various RE and EE solutions to further drive absorption of the offerings with these groups.

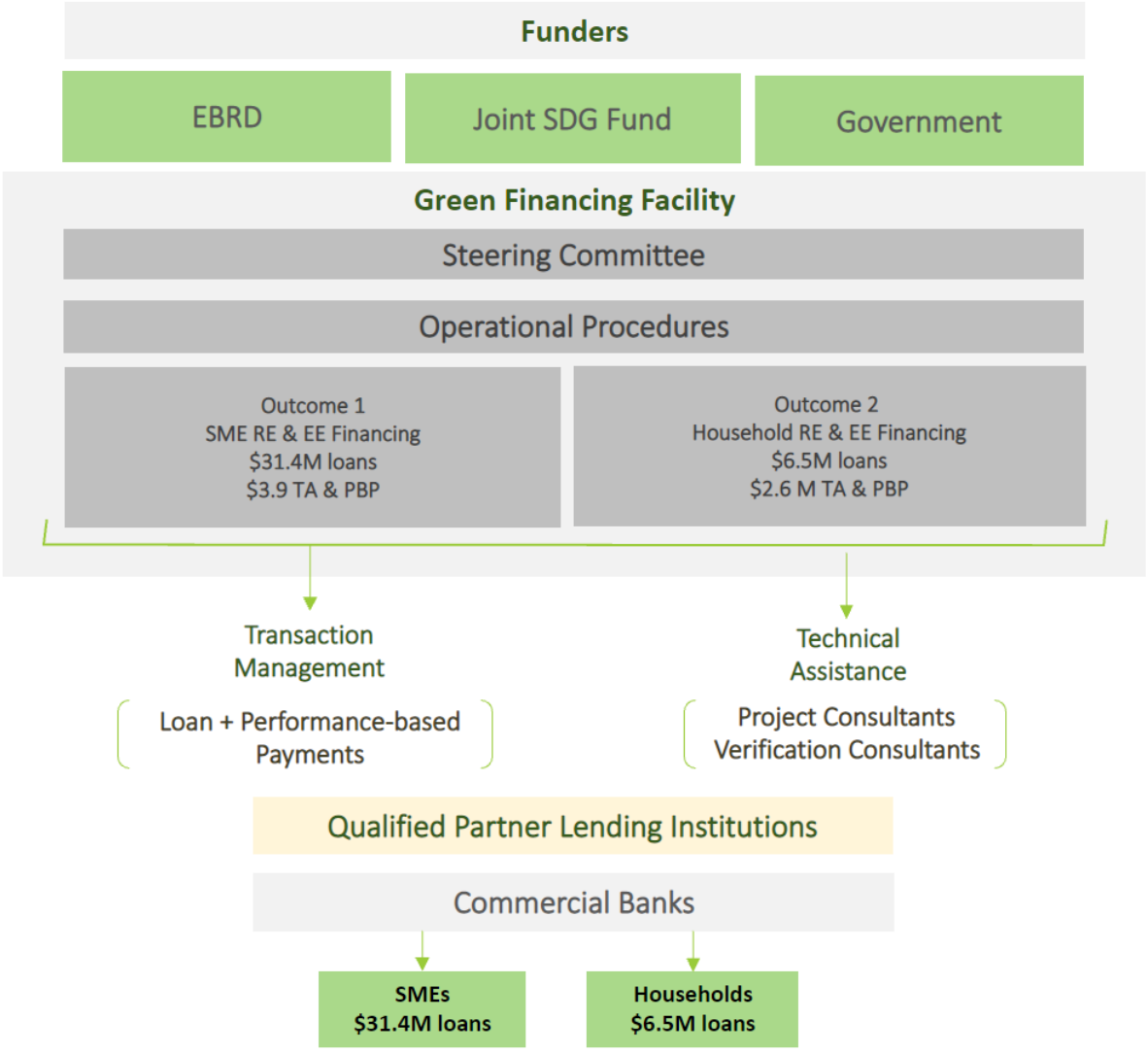
2.2. Proposed intervention

The project will establish a Green Financing Facility (GFF) as a flexible mechanism that builds on the comparative advantages, best practices, knowledge, and experience of all partners and offers comprehensive financing solution for accelerating a more rapid and wider adoption of RE and EE technologies by SMEs and targeted households. It will be implemented through at least **8 out of 13 banks in North Macedonia**, who together hold **90% of the market**, ensuring wide market outreach. The project consists of Outcome 1. Business RE/EE Financing and Outcome 2. Retail RE/EE Financing. A graphical representation of the financial structure of the project and at the outcome levels are presented below. Additionally, a financial and economic analysis of the project is presented in Annex 2.

The Theory of Change Diagram is presented in Annex 13.

[17] *More details on the engagement of local organizations that will work with project target groups is provided in the Communication Plan.*

Financial Structure



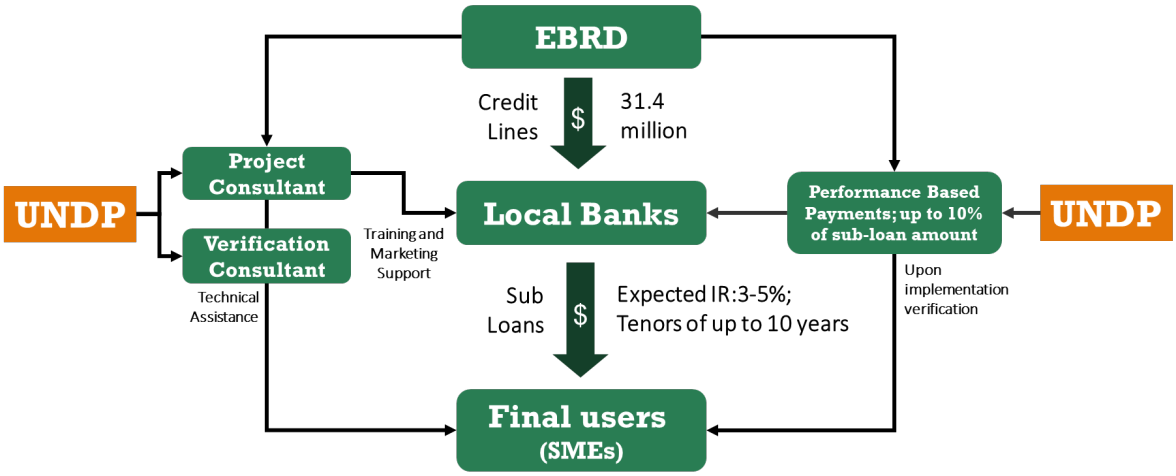
	Outcome 1: SME loan program	Outcome 2: Household loan program
Transaction Management	EBRD	UNDP
Technical Assistance	EBRD, UNECE	UNDP, IOM

The GFF will allow for efficient and reduced risk investing in RE and EE technologies via local markets/actors to the consumers. The GFF reduces investors' risks via the selection of verified commercial banks and associated creditworthiness check (reduced credit risk), verified project and verification consultants to ensure reliable RE/EE technical solution design (maximizing environmental outcomes and ensuring the funds will be spent by clients from the designated categories), transaction management (reducing transaction cost), a steering structure to ensure government and investors ownership and strategic decision making, backed by a monitoring and reporting system. At the same time, it is designed as a lean financing structure (a steering mechanism, a set of rules, transactions and legal arrangements) in order to keep operational costs to a minimum. To incentivize borrowers, the loan instruments will be subsidized via ex-post performance-based grants and free technical project design and verification services.

<u>Outcome 1:</u> SMEs have greater access to finance for accelerating a transition to renewable energy and energy efficiency solutions.

Outcome 1 focuses on the setup and implementation of Business RE/EE Financing. The project's setup activities for Outcome 1 include defining the rules and procedures, setting up sub-lending arrangements and documentation with local banks (UNDP and EBRD), and providing training for project and verification consultants (UNECE)[18]. Once the GFF Outcome 1 is launched - the project will provide monitoring and reporting services and administer the performance-based grants relying on its partners (EBRD) for transaction management. Based on the lessons learned from implementation and market demand, the project will invest in adjusting and standardizing the GFF rules and procedures (UNDP and EBRD) as well as in promoting the GFF to other investors.

Structure of Outcome 1



[18] *$101,520 will be provided to UNECE to design and implement trainings that will be tailored to support the consultancy companies in using the updated list of eligible technologies to design projects and monitor and verify the projects' performance (planned vs. actual). Also, UNECE will support the design of a system to measure the individual borrower's projects' impact in terms of their contribution to the overall project goals.*

Output 1.1: Finance allocated to banks and borrowers.

EBRD will provide USD equivalent[19] of EUR 27M ($31.4M as per UN Oct 2021 exchange rate) in the form of credit lines to local banks who on-lend these funds to SMEs in the form of sub-loans for investments into renewable energy and energy efficiency solutions that are in-line with the Policy Statement (set of rules) that will be established for the Facility. The average loan is estimated at EUR 0.3M while it will be capped at EUR 1M.

The participating local banks will be top tier banks – existing EBRD clients: Komercijalna Banka, NLB Bank Skopje, Sparkasse Bank Macedonia, Stopanska Bank and Procredit Bank Macedonia. These banks have a large market share **(75% in total),** access to the largest number of SME clients and vast experience in grant supported funding, as they are past participants under such schemes conducted by EBRD. Accordingly, these banks are best positioned to rapidly and effectively implement this new facility and deploy the loans quickly and effectively. Additionally, these banks will provide co-financing by sponsoring design and implementation of marketing activities of the offer, as well as the value and importance of RE and EE solutions. These five banks have submitted Letters of Interest, expressing their readiness to join GFF (see attached Letters at prodoc beginning).

Output 1.2: Performance-based payments deployed to qualifying commercial projects

The UNDP/Joint SDG Fund will provide $ 3.14M that will be allocated to loan recipients in the form of ex-post performance-based payments (PBP) for installed projects that have been externally verified to meet the established and agreed investment criteria.

PBP will range in value from 5% to 10% of the loan value.

Applications that only include EE will receive a PBP of 5% of the loan value while those in RE and those combining both RE and EE will receive a performance-based grant of 10%.

The rates of concessionality are determined based on the past market experience and lessons learned from EBRD's previous programme for supporting SME EE – WeBSEFF (the funds were fully on-lent by end of 2018) as well as analysis that confirm barriers for RE are higher[20].

The market study[21]confirmed that since the market has not changed significantly from the time that WeBSEFF II funds expired (2018), local banks expect that the same level of grants would result in the same or perhaps even a higher level of interest from consumers for such loans as prices of energy have risen since 2018. Equally, the study has shown that final beneficiaries find the grant level adequate as it significantly reduces the amount of money to be paid in interest.

Output 1.3: Projects designed and verified with technical assistance

The UNDP/Joint SDG Fund is to provide $ 1.57M for technical assistance via the GFF. The technical assistance will be delivered by two separate groups of consultants – Project

[19] *Based on the UN exchange rate of 1 EUR = 1.16 $(as at 01. October 2021). For clarity, the EBRD will provide EUR 27M while: a) the $equivalent of 10% of this amount will be allocated for performance-based payments and b) the $equivalent of the 5 % of this amount will be allocated for technical assistance. Note that "$" refers to USD.*

[20] *"De-risking investments in North Macedonia" IRENA 2021.*

[21] *Refer to Annex 10. Market study for details on aim, scope and methodology of the study*

Consultants (PCs) and Verification Consultants (VCs), which will work with the banks and with clients. Both groups will be trained and provided with advisory support on the design and verification of RE and EE projects.

Project Consultants will provide technical assistance to the banks and loan recipients supporting the project design process to ensure that projects align with the investment criteria. A technology selector tool (based on EBRD's existing tool) will be developed as part of the GFF efforts to maximize environmental and economic outcomes.

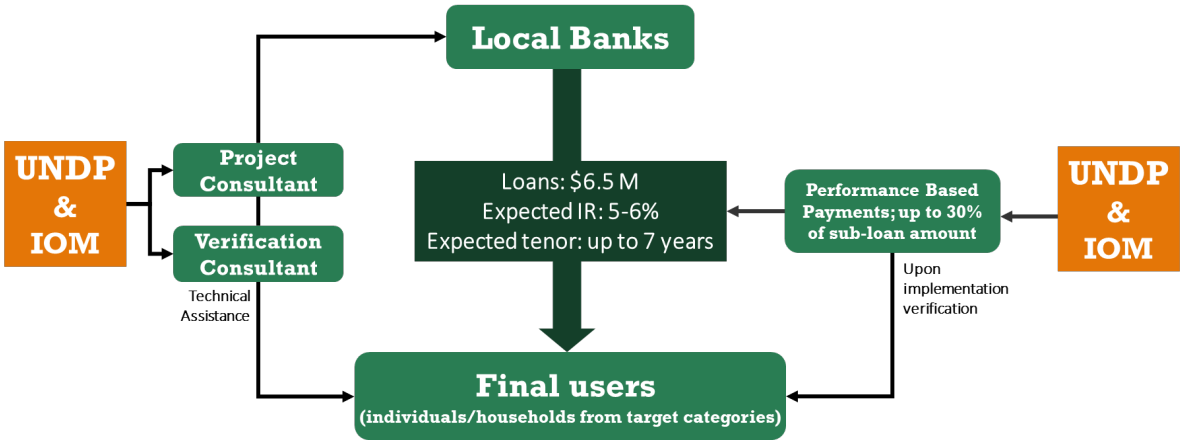
The PCs will be the backbone of project implementation and will be responsible for the following areas: a) **Pipeline development and structuring support:** Assist banks to develop a pipeline of qualifying projects by providing potential borrowers with consultation services to identify their specific RE and EE investment requirements and to support them in designing relevant solutions; b) **Supporting the project lifecycle:** Ensure that eligibility is determined in-line with the GFF Policy Statement; prepare project assessment documents/investment plans that will explain the benefits and savings of the underlying RE/EE investment, and assist the borrowers with the formulation of loan applications; c) **Capacity building/ initial training:** Train loan officers at banks to ensure that the technical criteria of the Facility are clearly defined in all project documentation, are consistently applied, and are updated throughout the life of the Facility. Loan officers will also be trained to collect and analyse sex-disaggregated data to support evidence-based design of financial products and decision making; d) **Marketing and awareness-raising:** Provide assistance in marketing the Facility to ensure that the widest range of stakeholders are informed about the offer and its benefits driving uptake. The outreach will utilize chambers and CSOs to ensure businesses from underrepresented groups including women-owned businesses are reached; and e) **Monitoring:** Ensure consistent communication among the banks, project developers, the Verification Consultants, and other stakeholders, and establishes an efficient tracking, monitoring and reporting system to ensure accurate data and standard forms are utilized including those facilitating sex-disaggregated data collection; and **Verification Consultants** will ensure that funded investments have been completed according to the original investment plans. Performance-based payments to sub-borrowers are only paid following successful verification by selected verification consultants of the completed Sub-projects.

Outcome 2:

Target groups/households, especially women and female headed households have greater access to finance for accelerating a transition to renewable energy and energy efficiency solutions.

Outcome 2 focuses on the setup and implementation of GFF Retail RE/EE Financing and activities which are similar to the ones developed under Outcome 1, albeit with a different source of finance, different local bank partners, different level of concessionality and greater promotion and advisory support to borrowers. Outcome 2 activities include defining the rules and procedures, setting up sub-lending arrangements and documentation with local banks (UNDP and IOM), and providing training for project and verification consultants. Once the Outcome 2 is launched, the project will provide monitoring and reporting services and administer the performance-based grants (UNDP and IOM). Outcome 2 specifically targets female-headed households, single-parents, households with PWD, Roma households, and employees who were affected by COVID-19. To tap into remittances, the project also targets remittance recipients and returning migrants.

Structure of Outcome 2



Output 2.1: Loans allocated to borrowers

Three local mid-tier banks, Halkbank, Silk Road Bank and Stopanska Banka Bitola will provide the equivalent of $ 6.5M of loans to targeted borrowers to finance investments in RE and EE solutions. Borrowers will be provided with a list of eligible equipment that meets the necessary energy generation and savings criteria as well as suppliers who sell such equipment. This list will be developed based on lessons learned from EBRD's Technology Selector for the ongoing Green Energy Financing Facility.

Participating local banks are smaller compared to those in Outcome 1, but with solid performance, have wide network of branches throughout North Macedonia (15-20 each, in most cities in the country), are retail orientated (at least 50% of their portfolios are retail), have adequate credit risk management standards (their levels of non-performing loans are around the market average) and also have experience with green financial products. They are eager to earn market share through the introduction of new products and have a higher risk appetite. Given all this, they are considered to be the ideal candidates for cooperation in the implementation of this Outcome. Additionally, these banks will conduct marketing activities of the offer, as well as promote the value and importance of RE and EE solutions. Halkbank, Silk Road Bank and Stopanska Banka Bitola have submitted a Letter of Interest, expressing their readiness to join Outcome 2 of the GFF, once available on the market.

The average loan size is expected to be USD 11,600[22]. The list of technologies eligible for the investment in EE and RE will be detailed in the list of eligible equipment that will build on the EBRD's Technology selector: http://technology-mk.ebrdgeff.com/. An average investment mix could be: $ 4,500 for a solar PVs for 4KWp of solar power; $ 2,500 insulation; $ 1,500 EE home appliances, and $ 1,500 EE heating appliances.

Output 2.2: Performance-based payments deployed to qualifying retail projects.

The UNDP and IOM will use the Joint SDG Fund grant to provide $1.95M for PBP to borrowers for installed projects that have been externally verified to meet the established and agreed investment criteria. In an effort to tap into the remittance revenues the programme aims to

[22] *US dollar amount calculated using the UN Exchange Rate at 01.Oct 2021 of 1 Euro = USD 1.16*

extend $0.45M out of $1.95M to remittance recipients and returning migrants. However, if issues with absorption are faced there is flexibility in the facility to reallocate to the other groups specifically targeted by the Outcome 2.

PBP will be calculated as a percentage of the total investment and will be 30% of the sub-loan amount. This concessionality rate is defined as slightly higher than the currently applied concessionality rate available on the market via the current Green Economy Financing Facility (GEFF) programme of 15-20%. The additional 10% top-up on the going market concessionality rate is added to ensure adequate incentivization of the GFF target groups, which are less affluent than the GEFF. The need for higher rate was confirmed by financial analysis of IRRs for sample investments for households (that indicated much lower values compared to those for businesses for the same level of subsidy), discussions with the banks and focus groups with households. The grant element should be higher to reduce the costs for the final users and reduce the risk of lending to the target groups.

All eight banks interviewed in the market study confirmed that a 30% PBP will be sufficient to incentivize the targeted underserved groups and will expand the client base of the banks. Also, the banks noted that they do expect that there will be sufficient demand within the target group (single parents, Roma population and Covid-affected citizens) to fully utilize the available funds. On the end-user side, this grant level will boost interest for borrowing among the households. Without the incentive the households are not likely to take the loan. 64% of the households surveyed in the market study confirmed that PBP-supported loans are the preferred source of financing after personal savings and public subsidy programmes. It needs to be noted that public subsidies are scarce, do not meet the market demand for EE/RE investment and are not targeting specific groups as they are on first come-first served basis.

Output 2.3: Projects designed and verified with technical assistance

The UNDP will use the Joint SDG Fund grant to provide $0.5M to engage consultants to provide technical assistance to the banks and loan borrowers to design qualifying projects and conduct post-implementation verifications. Also, IOM will use $0.15M from the Joint SDG Fund grant to specifically support returning migrants and remittance recipients. As in Outcome 1, there will be two separate sets of consultants – Project Consultants (PCs) and Verification Consultants (VCs). The PCs and VCs for Outcome 2 will be different from those engaged in Outcome 1 and will be engaged via different procurement processes.

The **Project Consultants** will provide technical assistance to the banks and loan recipients (borrowers) to support the project design process and ensure that projects are designed in line with project criteria/contributing to project outcomes and using the best technologies available at the market. The PCs will be responsible for the same delivery as in Outcome 1 but with households as the end-borrower. The **Verification Consultants** will ensure that the projects financed by loans have been completed substantially on the basis of original investment plans meet the eligibility criteria and have been used by users from the designated underserved groups. Performance-based payments will only be paid to borrowers after successful verification of the completed project.

UNDP and IOM will also engage in and support the promotion of the project offer to further increase the awareness and sensitize target groups about the value of RE and EE solution.

2.3 Leverage potential

The GFF has the potential to attract and mobilize capital at scale from both the private sector and public institutions as it provides a lean model of loans and incentives for investing in RE and EE solutions. So far, the project has secured co-financing from the EBRD and 8 private banks, which represents 90% of the overall banking sector as partners to the GFF.

	SDG Fund (in USD)	EBRD (in USD)	Local Banks (in USD)	Government (in USD)	Leverage (in USD)
Outcome 1	3,910,000	31,400,000		800,000	**8.2 : 1**
Outcome 2	2,600,000		6,500,000		**2.5 : 1**
Programme Mgt.	1,290,000				
Total	7,800,000	31,400,000	6,500,000	800,000	**5 : 1**

Overall Co-finance Leverage

Overall leverage of Joint SDG Funds in this project stands at **5 : 1** or **$ 38.7M : $ 7.8M.** Specifically, with $ 7.8M funding from the Joint SDG Fund, $ 31.4M is mobilized from EBRD, $ 6.5M from local banks and $ 0.8M from the government, totaling $ 38.7M. Letters of Intent/of Support confirming the pledge of these co-financing contributions have been secured from all partners (EBRD, 8 banks, and the Government of North Macedonia and all letters are attached to this Project Document).

In Outcome 1 the SDG Fund contributions of $ 3.9M will leverage $ 32.2M (IFI co-funding of USD equivalent of EUR 27M, i.e., $ 31.4M using UN Oct 2021 exchange rate and government co-funding of $ 0.8M). This gives a leverage of Joint SDG Funds of 8.2 : 1.

In Outcome 2, co-finance provided by local financial institutions in the form of loans to households will be at $ 6.5M enabled by $ 2.6M in funding from the SDG Fund, meaning that the leverage for this Outcome stands at 2.5.

Examining leverage at the Outcome level, Outcome 1 provides a higher level of leverage compared to Outcome 2. This is due to the higher incentive and technical support needed to attract households/individual clients (from the marketable but underserved target groups), compared to SMEs.

Private Finance Leverage

Overall private finance leverage stands at **4.9 : 1** i.e., **$ 37.9M : $ 7.8M** (IFI and private sector partner co-financing raised vis-a-vis Joint SDG Fund financing).

In Outcome 1, $ 3.9M unlocks IFI (EBRD) funding of EUR 27M[23] /$ 31.4M. Therefore, private finance leverage under Outcome 1 is 8 : 1.

[23] *Based on an exchange rate of 1 EUR = 1.16 USD the total value of EBRD's loan value of EUR 27M is equivalent to USD 31.4M.*

In Outcome 2, a total of $ 1.95M performance-based payments will be allocated to households/individuals while $ 0.65M will be invested in TA totaling in an investment of $ 2.6M from the SDG Fund. The value of performance-based payments will be up to 30% of the value of the loans - meaning that $6.5M will be coming from banks as loans. Therefore, leverage for this Outcome is 2.5 : 1.

2.4 Value added

Concessional funding from the Joint SDG Fund is required to incentivise SMEs and households to prioritize investments in RE and EE technologies. Banks have not provided this offering, that focuses on RE and EE solutions only, as the demand is insufficient without incentives and technical assistance. Responses from the banks (8 leading commercial banks and the country's Development Bank) and interviewed end-borrowers[24] confirmed that PBPs arethe most effective concessional financial product for the Macedonian context. For the FIs, it is a mechanism that attracts clients. For the final beneficiaries, it is a mechanism that provides the necessary impetus to undertake investments, in this case in EE/RE.

The market analysis[25] also points out that PBP-supported funding is the familiar instrument on the market and the one with which both the banks and the final users have the best experience in terms of improving the investment parameters. PBPs directly improve financial indicators as they reduce financing costs. This is the main motivating factor for the clients to prioritize certain investments. Hence, it is not surprising that the study demonstrated that 96% of SMEs prefer subsidized loans (loans with grant component based on verifiable achievement of EE/RE investment related benefits) over other instruments, including risk-sharing schemes.

Experiences with risk-sharing schemes have not been positive in North Macedonia. Risk-sharing schemes are viewed by the final users as an indirect support instrument that mostly benefits the FIs while not providing sufficient financial incentive to the final beneficiaries[26]. Furthermore, the market study has shown that risk-sharing instruments help local FIs to improve their portfolio risk profile. However, the scope of benefits passed on to the final beneficiaries depends on the risk perception of each FI individually, which is not a tangible criteria on which the final borrowers can rely. It happens quite often that the risk-sharing has only a small influence on the pricing as most of the reduced risk is materialized through easing the approval criteria, reducing the collateral requirements and other benefits that in Macedonian context are clearly perceived as less attractive than PBPs by the borrowers.

Additional argument in favour of the proposed GFF structure is the availability of TA. These investments require greater understanding of the RE and EE technologies that the banks do not have, so they perceive higher risk of placement, successful monitoring and implementation of these loans. The additional expert support provided via the project including the verification consultants, mitigates their perceived risk of entering in these new loan products.
Hence, GFF is a unique solution that is a tailored, transparent local financing mechanism that bridges the gaps, while also subsequently serving as a tool for the government, development partners, climate funds, investors and others to channel finance into green investments and

[24] *Refer to Annex 10. Market study for details on aim, scope and methodology of the study*

[25] *Refer to Annex 10. Market study for details on aim, scope and methodology of the study*

[26] *Risk-sharing instruments in North Macedonia and in the Western Balkan region were traditionally reserved for large infrastructure finance and general municipal lending projects offered by the country's Development Bank or major IFIs such as EBRD or KfW.*

a greener (COVID-19) recovery. GFF will be the first facility in North Macedonia to put emphasis on the RE investments, particularly on investments in solar panels, which were neglected by the local SMEs so far, as investments in new equipment usually have the priority. Only about 9% of the previously available SME loan financing for RE and EE were placed into RE projects (and none in photovoltaics) while about 7% of the currently available SME Competitiveness Programme (which has wide usage, and offers possibility for RE investments) were placed in such investments.

2.5 Innovative nature

Outcome 1 will provide for the first time in North Macedonia a finance facility that will promote combined investments in both RE and EE, with an emphasis on solar RE investments. Previous facilities were concentrated predominantly on EE with support for RE investments restricted to small hydro-power plants which had to be a new business line in order to qualify for support. By limiting the sub-loan amount to EUR 1M, the GFF will promote RE investments by the SMEs for their existing lines of business. Banks and end-users welcomed the GFF focus on RE investments during the market research, given that the solar segment is still underdeveloped (only 2% of the national technical potential has been utilized so far), despite the natural/climate advantages that country possesses.

Outcome 2 serves important groups of stakeholders who have not been included in other support mechanisms. The participating banks confirmed that they do have clients from target groups, although they are not separated from the other clients. However, the possibility to include grant supported funding in their portfolios - currently the most popular type of product on the market but so far not available to these banks - does incentivise them to provide funding through the GFF to the underserved target groups. Innovative approaches to stimulating RE and EE investments among the underserved target categories of the population are needed given that around 75% of the total GHG emissions originate from the energy sector[27]. As indicated currently 32% of households have any insulation while utilization of the solar potential is still very low (2% of the national technical potential). In addition, project is aiming to incentivize the use of remittances for RE and EE solutions.

2.6 Results

How the outcomes relate

Outcome 1 and 2 are complementary as they both have a similar framework but with different groups of beneficiaries and banks. Outcome 1 supports SMEs and Outcome 2 supports households. In this manner, they provide and mobilize a significant amount of capital for RE and EE solutions creating positive environmental benefits and building the ecosystem for sustained investments and utilization of these solutions beyond the life of the project. Furthermore, both outcomes will collectively drive growth of the market and utilization of RE and EE solutions leading to compounding environmental, social and economic benefits.

Metrics to measure the contribution of the project to SDG acceleration:

Description	Indicator	Baseline	Project impact	Contri-bution	Unit

[27] *National GHG inventory report, 2020*

SDG Target 7.2: By 2030, increase substantially the share of RE in the global energy mix.	Contribution of the project to the increase in the installed RE capacity generation potential	Installed capacity in the country excluding big hydro (586 MW) is 194 MW (source Annual report 2020, E|R|C)	10.7 MW	5.5	%
	Contribution of the project to the increase in the installed solar capacity generation potential	Installed capacity in the country 30 MW (source Annual report 2020, ERC)	10.7 MW	35.7	%
	Contribution of the project to the increase in the renewable energy share in total energy consumption	Renewable energy generation in the country in 2020 - 1,498 MWh (source Annual report 2020, ERC)	16,052 MWh	1.1	%
SDG Target 7.3: By 2030, double the global rate of improvement in energy efficiency.	Contribution of the project to the annual final energy consumption reduction target	Annual reduction target for the country up to 2030 2,406,616 MWh (source NDC)	70,068 MWh	2.9	%
SDG Target 13.2: Integrate climate change measures into national policies, strategies and planning; Indicator 13.2.2: Total greenhouse gas emissions per year	Contribution of the project to the annual GHG emission targets (up to 2030)	GHG emission reduction target for the country up to 2030 - 3,328.82kt (source NDC figure 4)	80.51 kt	2.4	%
Additional indicators	Share the final energy consumption reduction target in the sector industry for 2030	Reduction target for the sector industry in 2030 - 922,538.5 MWh (source NDC)	81,000 MWh (Outcome 1 loans with PBP to SMEs)	8.8	%
	Share of the final energy consumption reduction target for all sectors in 2030	Reduction target for all sector in 2030 - 3,238,82 MWh (source NDC figure 4)	86,200 MWh (entire project)	2.7	%

Note that Outcome and Output level targets and indicators are presented in detail in Annex 3, item 3.2 Joint Programme Results Framework.

Beneficiaries

The direct beneficiaries of both Outcomes will be the participating SME's and individuals/households. The total number of these direct beneficiaries is anticipated at 105+ enterprises and 650+ households/individuals with $0.3M and $10k being the average loan size for Outcome 1 and Outcome 2 respectively. While the project has specific beneficiaries it is also foreseen that all 2M people living in North Macedonia will benefit indirectly from the reduction of emissions and cleaner air.

Female headed households, single parents, households with persons with disabilities (PWD), Roma households, and employees who were affected by COVID–19 are groups that the project specifically targets through Outcome 2. Existing social and economic inequalities are in direct correlation to an individual's ability to invest in replacing inefficient heating and cooling devices as well as mitigate the negative effects of climate change and air pollution. In that regard, women are more vulnerable across different groups due to gender norms, lower access to financial instruments, and lower participation in the labour market. The financial instruments of this project will proactively seek to inform and target women– specifically women-led companies, women entrepreneurs and female-headed households. The project also targets remittance recipients and returning migrants.

Alignment with National Policy

This project is aligned with North Macedonia's national goals as the country has committed to reducing its CO2 emissions by 51% compared to the 1990 level, as well as with the targets set in the new Strategy for Energy Development till 2040. The project shall also contribute to the implementation of the National Plan for Clean Air. North Macedonia is the first country in the region that has fully merged energy and climate policies, which also supports the process of accession to the European Union.

North Macedonia's strategies, plans and commitments are outlined in key documents. As of August 2021, the documents include climate commitments in the NDCs, the NSED, the NECP, the Renewable Energy Action Plan for the Republic of North Macedonia until 2025 with vision until 2030, and the Long-Term Strategy and Law on Climate Action until 2050. Additional strategies and plans are still awaiting formal government adoption to enter into force.

North Macedonia has initiated an ambitious decarbonization pathway that forms a solid basis for attracting renewable energy investments in the country. To date, such efforts have focused on renewable energy for power with some attention given to heating and cooling in the most recent strategy and plans.

Alignment to the UN Sustainable Development Cooperation

This project is aligned to the UNSDCF Output 3.1 as it provides for the Macedonian citizens to benefit from ambitious climate action by enabling access to finance for RE and EE solutions and by providing cleaner air than would be the case under Business-as-Usual scenario.

2.7 Gender and human rights plan

In developing this proposal, strong efforts were made to understand the barriers faced by women, women owned businesses, women-headed households and female remittance recipients, unique impacts and needs of target groups as it relates to the access to green finance and adoption of RE and EE products. The process to produce these Outcomes included a desktop review of available literature, analysis of relevant legislation, policy and practices of financial institutions in the country, as well as interviews and consultations with project and beneficiary stakeholders.

Despite existing policies and measures to limit gender-discrimination and promote gender sensitivity and women empowerment, de facto historical and cultural factors contribute to enduring gender inequality in employment and remuneration, poverty, education and safety/security. The negative implications of gender inequality are anticipated to worsen under the impacts of climate change and the COVID-19 pandemic.

To mitigate against biasing against women and unintentionally contributing to the perpetuating or worsening gender inequality and gender-based vulnerability, the project seeks to promote gender equality and empowerment of women by including activities which will increase women's representation and awareness, as well as ensuring that any awareness and or training programs are gender-sensitive and inclusive.

In order to engage women as business owners and female led households, the project will partner with business associations representing women in business and women CSOs to ensure wide outreach and information distribution. In addition, women beneficiaries EE will be featured in promotional materials and outreach activities.

Research shows that women in business lack information about financial mechanisms and support and that they need financial literacy skills in order to utilize existing offers. Therefore, the project will focus on outreach to women and will train loan officers to have a better understanding of women in business and women led households.

More broadly, UNDP is presently supporting the Ministry of Environment and the Ministry of Labour and Social Policy to integrate gender into climate change policies, via the Gender and Climate Change Action Plan. UNDP will continue supporting the government and the CSOs as duty bearers in increasing their capacity to integrate gender into climate change over the long-term by providing evidence from project results and activities. Financial mechanisms that scale up the adoption of renewable energy by households and individuals can have a sustainable impact on the vulnerable. CSOs are some of the rights holders' organizations that will be involved in the outreach activities and in tracking the project results in order to support scaling-up.

Close cooperation between the duty bearers and the rights holders has been established and will be integrated into the oversight mechanisms. Implementation partners were identified and selected based on how they complement each other with diverse strengths but also share common values around the promotion of human rights and gender equality. In reviewing the recommendations issued to the country by human rights mechanisms, it was evident that there are no specific recommendations which relate to climate change and facilitating access to finance by women, including female remittance recipients and returning migrants. An ongoing IOM project in North Macedonia adopts a system's approach analyze the interlinkages between human mobility, air pollution and clean energy, especially in urban and peri-urban areas, including women and Roma. The findings of this project will inform the GFF and its stakeholders.

List of underserved groups	Dedicated Outcome	Dedicated Output
Women and girls	X	
Persons with disabilities		X
Minorities (Roma)		X
Migrants		X
Other groups: Remittance recipients, and female-headed households, single-parents and employees affected by COVID-19.		X

2.8 Progress

Outcomes 1 and 2 benefit from a substantial amount of preparatory work having been done to date that enables the allocation of loans to beneficiaries within 3-6 months of the project launch. There have been several rounds of consultations with the selected partner banks who

have committed to participate in the project (Letters with Expression of Interest from Local Banks are attached). The basis for the procedures, document templates as well as contacts with the network of the technical consultants (project and verification) is based on EBRD's positive experience with the implementation of similar credit financing facilities on the local and regional market, the Western Balkans Sustainable Energy Financing Facility (WebSEFF) which targeted local SMEs, and the Green Economy Financing Facility (GEFF) which targeted individuals. The abundance of EBRD experience in developing such grant supported financial schemes allows for a short period of adaptation and quick rollout to the market[28]. Operationally, the proposed structure of a steering committee and operational mechanisms have been discussed and agreed to by the key partners.

2.9 Sustainability

The project has been designed to deliver benefits beyond the lifetime of the project through the establishment of a model that can be utilized for providing access to finance for RE and EE solutions.

The sustainability of project will be further enhanced through the project's focus on ensuring more gender-equal benefits to project beneficiaries. One such benefit will be the enhanced capacity of loan officers to collect and apply sex-disaggregated data that will create basis for better understanding of gaps and barriers and subsequent better design of relevant policies and actions. Sex-disaggregated data is the basis for any gender analysis and this data can potentially provide fertile ground for modeling a cross-sectoral approach in developing financial instruments, measures and/or subsidies for RE and EE for target groups. Additionally, it will be beneficial for financial institutions to monitor the absorption of loans and contribute to increasing their internal knowledge management systems.

Upon project completion, it is expected financial services and the market for RE/EE solutions will be enhanced. With a larger and more active market - new market entrants such as product vendors, service providers and financial solutions are expected. The increased level of supply and demand should drive competition resulting in more affordable products, lowering payback periods, and reducing or eliminating the need for concessionality. It is expected that some of the participating banks will continue to service this market segment and this coincides with the conclusions partly drawn from EBRD's lessons learned from the WeBSEFF program, which showed that once the product is well accepted and widely known on the market that demand continues and increases.

2.10 Replicability

The project has been designed with the view for scaling and replicating across the Western Balkans and even to other regions, post a successful implementation. The project is expected to demonstrate the effectiveness of the Green Financing Facility model in North Macedonia which serve as a use-case and model to governments, financial institutions, investors, and others in the region on how to replicate in different countries with similar economies.

The project aims to catalyze impact beyond a one-off project investment using a combination of market push and pull activities that address barriers to the adoption of renewable energy and energy-efficient solutions. Ultimately the project should drive a paradigm shift through the establishment of a mechanism and a standard for financing and adoption of RE and EE solutions.

[28] *EBRD will require 3-4 moths for the internal approval process if/when the Joint Programme is approved by the SDG Funds.*

Key learnings will be the level of concessionality, loan rates, and value of technical assistance as well as the choice of bank partners. These will be captured through the quarterly reports to be received from the participating banks, but also through the regular quarterly reporting and the final reports to be prepared by both the project and the verifying consultants. A successful implementation will enhance the likelihood of securing finance from government budgets or other international institutions, such as the Green Climate Fund, whose mandate is to support mitigation efforts.

EBRD, UNDP, and IOM are capable, well-positioned, eager and committed to collaborate and scale up this effort to cover the entire Western Balkans sub-region and potentially other regions.

3. Programme implementation

3.1 Governance and implementation arrangements

A **Green Financing Facility Steering Committee** composed of the Deputy Prime Minister for Economic Affairs (chair), UNDP, EBRD, IOM, RCO, and any additional, future GFF Investors will be established and will meet on a semi-annual basis to review progress and provide strategic advice on project implementation. Minister of Finance, Minister of Economy and Minister of Environment and Spatial Planning will participate as observers in the GFF Steering Committee. Executive Boards per outcome can be established by parties engaged for more direct oversight of the implementation.

UNDP is the Lead Agency of the project and will oversee overall project implementation.

IOM is a partner agency of the project and will implement the parts of Outcome 2 that are focused on increasing the adoption of RE and EE solutions among returning migrants and remittance recipients.

UNECE is a partner agency and will provide technical assistance and capacity building on RE and EE technologies for SMEs to the project and verification consultants, in the first year only.

RC/RCO will provide support to the PUNOs including partnership building, financing, monitoring and reporting, and other coordination activities, as needed.

3.2 Partnerships and stakeholder engagement

The Deputy Prime Minister for Economic Affairs has been coordinating the Government's involvement in the preparation of the project, review by relevant Ministries and subsequent endorsement of the project at the Government session based on which endorsement letter has been signed. The Deputy Prime Minister for Economic Affairs will also be chairing the Steering Committee (as described in the Governance section) and facilitate the sharing of project results and lessons learnt by presenting them to the **National Council for Sustainable Development** – the highest national body overseeing sustainable development processes. Additionally, the relevant **Sector Working Groups**[29] (e.g., the national aid and sector policy coordination system, including government, civil sector, academia, donors and IFIs) will be informed on developments in the GFF. The use of the existing National Council and the Sector Working Groups as platforms for informing and discussing will reduce

[29] *Currently the relevant Sector Working Groups are the SWG on Environment and the SWG on Competitiveness and Innovation.*

transactions in ensuring wider coordination with key national stakeholders (e.g., government, CSO, IFIs, academia, etc.). If additional national coordination bodies specific to the themes of RE and EE are created, the project will seek participation in these to ensure greater use of national systems and better coordination.

The **involvement of the private sector** is primarily centered on the commercial banks in North Macedonia as they are a conduit between the GFF and final users of RE and EE solutions. **Eight out of the thirteen banks in North Macedonia** have expressed interest in participating in the project, allowing GFF to tap into a significant share of the market (the local banks that expressed interest have **more than 90% of market coverage**). The local banks were consulted in the preparation phase and their feedback with regards to product design, organization of consultants and their relations with the banks and the final users, and streamlining the processes, are the backbone for designing the GFF features. The local banks will be delivering the GFF loans to the final clients (SMEs and households). They will also provide information and specialized advisory services on RE and EE to clients through information campaigns, which they will run to promote the subsidized loans to clients and through the information on GFF approved RE and EE technologies and the project and verification consultants.

Given **EBRD**'s long track record in North Macedonia it has strong well-established relationships with both the top-tier and most prominent Macedonian banks. These relationships, accumulated knowledge and expertise together with UNDP's experience in brokering and managing complex projects, tacking complex development challenges, as well as IOM's specific expertise for addressing challenges among returning migrants and remittance recipients in the context of climate change and environmental degradation, are all of great value and importance to the establishment of this GFF and successful implementation of the project.

Stakeholder name	Role of stakeholder in structure	Level of engagement/support to date
EBRD	Project partner	Maximum level of engagement.
IOM	Project partner	Maximum level of engagement
Local Banks	Partner banks in Outcome 1 for lending to SMEs	Significant amount of engagement and a well-established relationship with EBRD.
Local Banks (other)	Partner banks in Outcome 2 for lending to households.	Significant amount of engagement and good knowledge of similar products offered by EBRD.
Vendors of RE and EE products and solutions	Key participants in the value chain.	Modest level of engagement as is appropriate.
Deputy Prime Minister	Key Government partner	Co-financing of $0.8M, endorsement letter and coordination on Government's behalf

3.3 Monitoring, reporting, and evaluation

Reporting on the Joint SDG Fund will be results-oriented, and evidence based. Each PUNO will provide the Convening/Lead Agent with the following narrative reports prepared in accordance with instructions and templates developed by the Joint SDG Fund Secretariat:

- *Annual narrative progress reports*, to be provided no later than one month (31 January) after the end of the calendar year, and must include the result matrix, updated risk log, and anticipated expenditures and results for the next 12-month funding period;

- *Mid-term progress review report* to be submitted halfway through the implementation of Joint Programme[30]; and
- *Final consolidated narrative report*, after the completion of the joint programme, to be provided no later than two months after the operational closure of the activities of the joint programme.

The Convening/Lead Agent will compile the narrative reports of PUNOs and submit a consolidated report to the Joint SDG Fund Secretariat, through the Resident Coordinator.

The Resident Coordinator will be required to monitor the implementation of the joint programme, with the involvement of the Joint SDG Fund Secretariat to which it must submit data and information when requested. As a minimum, joint programmes will prepare, and submit to the Joint SDG Fund Secretariat, 6-month monitoring updates. Additional insights (such as policy papers, value for money analysis, case studies, infographics, blogs) might need to be provided, per request of the Joint SDG Fund Secretariat. The Joint Programme will allocate resources for monitoring and evaluation in the budget.

Data for all indicators of the results framework, which will be shared with the Fund Secretariat on a regular basis, in order to allow the Fund Secretariat to aggregate results at the global level and integrate findings into reporting on the progress of the Joint SDG Fund.

PUNOs will be required to include information on complementary funding received from other sources (both UN cost sharing, and external sources of funding) for the activities supported by the Fund, including in-kind contributions and/or South-South Cooperation initiatives, in the reporting done throughout the year.

PUNOs at the Headquarters level shall provide the Administrative Agent with the following statements and reports prepared in accordance with its accounting and reporting procedures, consolidate the financial reports, as follows:

- *Annual financial reports as of 31st December each year with respect to the funds disbursed to it from the Joint SDG Fund Account, to be provided no later than four months after the end of the applicable reporting period; and*
- *A final financial report, after the completion of the activities financed by the Joint SDG Fund and including the final year of the activities, to be provided no later than 30 April of the year following the operational closing of the project activities.*

In addition, regular updates on financial delivery might need to be provided, per the request of the Fund Secretariat.

After the competition of the joint programme, a final, *independent and gender-responsive*[31] *evaluation* will be organized by the Resident Coordinator. The cost needs to be budgeted, and in case there are no remaining funds at the end of the joint programme, it will be the responsibility of PUNOs to pay for the final, independent evaluation from their own resources.

The joint programme will be subjected to a joint final independent evaluation. It will be managed jointly by PUNOs as per the established process for independent evaluations, including the use of a joint evaluation steering group and dedicated evaluation managers not

[30] *This will be the basis for release of funding for the second year of implementation.*
[31] *How to manage a gender responsive evaluation, Evaluation handbook, UN Women, 2015*

involved in the implementation of the joint programme. The evaluations will follow the United Nations Evaluation Group's (UNEG) Norms and Standards for Evaluation in the UN System, using the guidance on Joint Evaluation and relevant UNDG guidance on evaluations. The management and implementation of the joint evaluation will have due regard to the evaluation policies of PUNOs to ensure the requirements of those policies are met and the evaluation is conducted with the use of appropriate guidance from PUNOs on joint evaluation. The evaluation process will be participatory and will involve all relevant programme's stakeholders and partners. Evaluation results will be disseminated amongst the government, development partners, civil society, and other stakeholders. A joint management response will be produced upon completion of the evaluation process and made publicly available on the evaluation platforms or similar of PUNOs. For additional details on how the project will be monitored and evaluated, see Annex 6: Monitoring and Evaluation Plan.

3.4 Accountability, financial management, and public disclosure

The Joint Programme will be using a pass-through fund management modality where UNDP Multi-Partner Trust Fund Office will act as the Administrative Agent (AA) under which the funds will be channeled for the Joint Programme through the AA. Each Participating UN Organization receiving funds through the pass-through has signed a standard Memorandum of Understanding with the AA.

Each Participating UN Organization (PUNO) shall assume full programmatic and financial accountability for the funds disbursed to it by the Administrative Agent of the Joint SDG Fund (Multi-Partner Trust Fund Office). Such funds will be administered by each UN Agency, Fund, and Programme in accordance with its own regulations, rules, directives and procedures. Each PUNO shall establish a separate ledger account for the receipt and administration of the funds disbursed to it by the Administrative Agent.

Indirect costs of the Participating Organizations recovered through programme support costs will be 7%. All other costs incurred by each PUNO in carrying out the activities for which it is responsible under the Fund will be recovered as direct costs.

Funding by the Joint SDG Fund will be provided on annual basis, upon successful performance of the joint programme.

Procedures on financial transfers, extensions, financial and operational closure, and related administrative issues are stipulated in the Operational Guidance of the Joint SDG Fund.

PUNOs and partners must comply with Joint SDG Fund brand guidelines, which includes information on donor visibility requirements.

Each PUNO will take appropriate measures to publicize the Joint SDG Fund and give due credit to the other PUNOs. All related publicity material, official notices, reports and publications, provided to the press or Fund beneficiaries, will acknowledge the role of the host Government, donors, PUNOs, the Administrative Agent, and any other relevant entities. In particular, the Administrative Agent will include and ensure due recognition of the role of each Participating Organization and partners in all external communications related to the Joint SDG Fund.

3.5 Legal context

Agency name: UNITED NATIONS DEVELOPMENT PROGRAMME (UNDP)
Agreement title: STANDARD BASIC ASSISTANCE AGREEMENT (SBAA)
Agreement date: New York, 18 October 1995 and Skopje, 30 October 1995

Agency name: INTERNATIONAL ORGANIZATION FOR MIGRATION (IOM)
Agreement title: MEMORANDUM OF UNDERSTANDING (MoU)
Agreement date: Geneva, 6 April 2001 and Skopje, 20 July 2001

The United Nations Economic Commission for Europe (UNECE) is part of the UN Secretariat and its mandate and activities are defined in the UNECE Terms of Reference and Rules of Procedure of the Economic Commission for Europe (E/ECE/778/Rev.5).

D. ANNEXES OF THE JOINT PROGRAMME TEMPLATE

Annex 1. List of related initiatives

The **World Bank and the Government** commenced the North Macedonia Public Sector Energy Efficiency Project in January (2020). This project aims to reduce energy consumption in the public sector and support the development and implementation of a sustainable financing mechanism for energy efficiency in the public sector. Together, the WB Public Sector Energy Efficiency Project and the GFF will result in a positive compounding effect that will lead to reductions in carbon emissions, better economics on the purchasing of energy, cleaner skies, and improved socio-economic conditions.

The **European Bank for Reconstruction and Development (EBRD)** provides credit and equity financing for the private sector. In terms of RE, EBRD's work is focused on large scale projects, while for EE, EBRD offers credits to the higher-end residential sector via the **Green Economy Financing Facility** (GEFF) - a concessionalised credit line distributed via local banks focused on the financing of energy-efficiency products. GEFF total volume for the Western Balkans is EUR 85 million, financed by the EU, Austria and the WBIF donors. EBRD addresses, inter alia, climate change mitigation, through the EBRD Green Cities Initiative, which includes the capital city of Skopje.

The proposed project is complementary to the GEFF, covering a different, largely underserved market segment of borrowers as GEFF focuses on more affluent early adopters. It complements the Green City Skopje initiative and potentially serves as one of the tools in addressing the city's climate change mitigation and adaptation efforts. Also, the experience, networks and setup of the GEFF have been used as basis for the development of the Outcome 2 Green Financing Facility.

Habitat for Humanity Macedonia (HH) offers favorable financial support for residential EE and home improvement. HH's main activity in this area is focused on financing for EE in multi-family residential buildings. This retail lending credit line is approximately 0.9M USD. HH also offers a guarantee fund of 0.1M USD to help cover losses by banks when extending EE loans as well as loans for the repair of houses in partnership with local savings houses Moznosti and Horizonti, targeting vulnerable groups.

Via the *Law for Financial Support for Investment* **the Government of North Macedonia** provides financial support to companies for investments that increase their competitiveness. The overall aim of this financing support is to stimulate economic development. The per-project ex-post grant targets up to 10% of the overall investment and ranges from EUR 0.4 to 1 million per-project. The funds are available for investment in any productive assets that increase the competitiveness of the company. At the household level, the Government also offers grants to support the purchase of pellet stoves, solar panels and insulation. Women are underrepresented in this scheme only receiving 26% of the allocated funds for pellets stoves, 25% for solar panels, and 27% for PVC and or window insulation. A major pitfall of this support programme is that it has been based on a "first-come, first-served" basis without identifying, targeting or encouraging disadvantaged and/or vulnerable groups to apply for these subsidies.

Name of initiative / project	Key expected results	Links to the joint programme	Current status	Lead Org	Other partners	Budget and funding source	Contract person (name and email)
North Macedonia Public Sector Energy Efficiency Project	EE Investments in Municipal Sector. EE Investments in Central Government Buildings. Technical studies to support investments. Technical assistance and implementation support to drive EE in the public sector. Initial capital to support EE Fund staff, operating costs, marketing, initial audits/designs and investments.	The WB project focuses on developing sustainable financing mechanisms for energy efficiency in public sector buildings while the JP focuses on the same but with private buildings. Together, these two projects will result in a positive compounding effect that will lead to reductions in carbon emissions, better economics on the purchasing of energy, cleaner skies, and improved socio-economic conditions	Loan approval pending; Implementation expected to start in 2021	World Bank	The Government of North Macedonia	EUR 25 M World Bank	Rhedon Begolli rbegolli@worldbank.org
Green Energy Financing Facility (GEFF)	Contribute towards building a green economy by facilitating the expansion of residential energy efficiency lending in North Macedonia and positive demonstration effects of green economy projects. GEFF is also contributing towards building a more competitive financial sector through developing the internal capacity of the PFIs for financing green economy projects.	Basis for design of the Outcome 2 of GFF.	Implementation started in 2017, EUR 21m committed to North Macedonia to date. GEFF II facility extension was introduced in 2021 with EU 8m signed to date by the Macedonian banks and an additional EUR 2m to be committed by year end.	EBRD	Ohridska Banka, Sparkasse Bank, Procredit Bank, and NLB Bank Skopje	EUR 21.5 m of credit lines / EBRD EUR 4.3 m of incentives / EU EUR 5.1m for TC / Gov of Austria and the EU	Naum Ribaroski ribarosn@ebrd.com Beata Paroczai ParoczaB@ebrd.com
Western Balkans	Demonstration effect of improved EE in North	Basis for design of the Outcome 1 of GFF.	Implementation lasted	EBRD	Ohridska Banka,	TC and grants	Naum Ribaroski; ribarosn@ebrd.com

Sustainable Energy Financing Facility (WeBSEFF)	Macedonia by establishing the benefits of energy conservation to a wide range of sub-borrowers. Transfer of expertise related to RE and EE investments among financial intermediaries and sub-borrowers, resulting in lowering the transaction costs for financing the sustainable energy investments.		from 2012 until 2018. EUR 24m of sub-loans have been placed to the final users through three local banks (Ohridska, NLB and Halkbank).		NLB Bank Skopje and Halkbank	funded by EU	Beata Paroczai; ParoczaB@ebrd.com
Habitat for Humanity Macedonia (HH) loan and guarantees	Favourable financial support for residential EE and home improvement. HH's main activity in this area is focused on financing for EE in multi-family residential buildings.	Exchange of information on the programme in the preparatory phase. Lessons learned integrated in the GFF design.	Active.	HH	Savings House Moznosit and Horizonti	A retail lending credit line with total volume of USD 0.9M and a guarantee fund of USD 0.1M.	Liljana Alceva, Deputy Director for Program lalceva@habitat.org.mk
Law for Financial Support for Investment	Provides financial support to companies for investments that increase their competitiveness. The overall aim of this financing support is to stimulate economic development.	Exchange of information on programme in the preparatory phase.	Active.	Cabinet of the Deputy Prime Minister for Economic Affairs	Directly provided by government entities to companies	EUR 0.4 to 1 million per-project, not specifically targeted to RE and EE	Maja Barik, Head of Unit for monitoring and analysis of state strategies, competitiveness, business climate and coordination of projects supported by the international community Cabinet of the Deputy Prime Minister of Economic Affairs Maja.barik@gs.gov.mk

Annex 2. Financial and Economic Analysis

The project team has modelled the financial returns for buyers of RE-EE equipment as a result of the electricity savings realized over the life of the equipment.

For SMEs (Outcome 1) the following assumptions were made:

- Three investment options: RE only (solar PVs), EE only (e.g. building insulation, energy-efficient factory equipment), and both RE-EE;
- Average loan size under each option derived from EBRD WeBSEFF prior loans: ~$53k for RE, $242k for RE and $295k for RE-EE;
- 20% down payment by SMEs; This is based on the previous experience with the WEBSEFF II SME EE Facility where the total down payments stood at 23% and market analysis that indicated similar down payments can be expected for GFF;
- Annual electricity savings based on prior WeBSEFF experience. Non-residential electricity tariff of $0.10/kWh as per latest national regulation;
- Financial incentive of 10% of loan value for RE, 5% for EE and 10% for RE-EE;
- Returns calculated under 4 loan scenarios: 5 and 10yr loans at 3% and 5% rate
- Other assumptions: 20yr lifetime of equipment; operations and maintenance (O&M) cost ranging from 1.7% p.a. for RE to 5% p.a. for EE (on total equipment value); 10% corporate tax rate.

Under these assumptions, the financial internal rate of return (FIRR) for SMEs, without concessionality (i.e., without financial incentive) would be: 17-27% for RE (depending on the loan scenarios), 9-17% for EE and 11-19% for RE-EE. Financial incentives (concessionality) would increase FIRRs to: 20-36% for RE, 11-21% for EE and 14-30% for RE-EE. While FIRRs would be relatively attractive already without concessionality, financial incentives are deemed essential for the successful penetration of RE and EE in light of the following:

- Incentives as a percentage of loan amount and overall investment (loan + down payment) are small;
- The market for RE-EE is still nascent in North Macedonia. The previous EBRD WeBSEFF windows financed 67 investments,[32] simply kick-starting the market, which remains very far from mature. The EBRD deems incentives essential to increase the penetration of RE-EE. This is particularly true for RE, which was not an area of focus of prior EBRD WeBSEFF windows;
- Financial incentives are needed for SMEs to prioritize investments into EE&RE and to reduce the upfront cash outflow (investment size less loan) for SMEs, making the investments more attractive.

For households (Outcome 2) the following assumptions were made:

[32] *Not counting 6 mini-hydro investments that would not be eligible under this project.*

- Average loan size of $10,000 to acquire solar PVs and insulate the house;
- 20% down payment by households; this estimate is based on the current experience with EBRD's GEFF programme, where average down payments currently stand at 25%.
- Financial incentive equivalent to 30% of loan amount;
- Four loan scenarios: 5 and 7yr loans at 5% and 6% rate
- 8,000 kWh of electricity saved annually (from solar PV and insulation), at the residential electricity tariff of $0.135/kWh as per latest regulation;
- Other assumptions: 20yr equipment life and modest O&M costs of 1% p.a.

Under these assumptions, the FIRR for a household, without concessionality, would be 4-5% (depending on loan scenario). The financial incentives would double FIRRs to 8-10%. While this level of returns is not as attractive as that for SMEs, it is important to note that the main investment driver for households, in addition to electricity savings, are health benefits, and increased comfort of the house through better insulation, windows etc. In addition, financial incentives for households are greater than the down payment and therefore significantly reduce the negative cashflows incurred by buyers upfront. In summary, incentives are expected to play a major role in prioritizing EE & RE investments, kick-starting and upscaling the retail RE-EE market.

For both households and SMEs, it is expected that over time, after the conclusion of the project, the need for financial incentives will drop or disappear, as buyers become more familiar with RE-EE investments due to the project interventions. This project, in conjunction with the similar EBRD projects on the market (GEFF, upcoming GEFF II and the SME Competitiveness Support Programme) will help create economies of scale to reduce equipment prices. In addition, with EU accession it is generally expected that electricity tariffs in North Macedonia will be on an upward trend. While the increase is not expected to be imminent – tariff increases are socially and politically sensitive – in the long run higher electricity prices will increase the FIRR of RE-EE investments, further reducing the need for incentives.

The economic analysis estimates the value of the societal benefits of the project vs. project cost, the latter including both the SDG Fund grant and co-finance. The two main quantifiable benefits of the project are:

- The electricity savings realized by all the SMEs and households that invested in RE-EE with project support (68,030 MWh p.a. at full regime) at current tariff levels[33]
- Reduction in CO2, estimated applying a conversion factor of 0.934 tCO2eq per MWh of electricity saved, for a total of 63,540 tCO2eq p.a. at full regime.

[33] *Both the economic and financial model are on a real basis, i.e. assume zero inflation.*

Estimates of carbon prices vary widely[34]. The World Bank recommends that economic analyses of projects use an estimated range of carbon price of USD 40-80 (min-max) in 2020, shifting to USD 50-100 (min-max) by 2030.[35] The European Investment Bank (EIB) and European Commission (EC) recommend using a central value of EUR 25/tCO2eq in 2010, raising gradually to EUR 45 by 2030.[36] These figures are considerably higher than the (admittedly limited) evidence from emission allowances traded on exchanges. As of November 2020, when the economic model was finalized, in the EU-ETS carbon exchange (the largest and most liquid), the price of CO2 allowances was approximately EUR 59 (USD 70) per ton. Conservatively, this analysis uses the EU-ETS price.

Based on these assumptions, the project will generate an economic IRR (EIRR) of 24% and economic net present value (NPV) of US$ 37 million.[37] These results are already attractive *per se,* even without considering the further economic benefits resulting from reduced pollution and improved health. Estimates of the latter would require detailed econometric studies. On a purely indicative basis, an EIRR sensitivity has been run based on the OECD's finding that 1µg/m3 reduction in PM2.5 concentration in Europe boosts GDP by 0.8%.[38] If the proposed project caused only 1/100 of that boost (0.008% of GDP), EIRR would increase to 28% and economic NPV to US$ 47 million, further strengthening the case.

[34] *The Draft Law of Climate Action provides legal basis for introduction of a CO2 tax. In addition, the National Climate Strategy foresees introduction of CO2 tax. The prices for the modeling work done in the framework of the Climate Strategy are taken from the World Energy Outlook and are in the range of the process recommended by the EIB.*
[35] *World Bank (12 November 2017). Shadow Price of Carbon in Economic Analysis – Guidance Note.*
[36] *European Commission (December 2014). Guide to Cost-Benefit Analysis of Investment Projects. Link: https://ec.europa.eu/regional_policy/sources/docgener/studies/pdf/cba_guide.pdf*
[37] *With a 10% discount rate used by UNDP and other international organizations in economic analyses.*
[38] *http://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?cote=ECO/WKP(2019)54&docLanguage=En*

Annex 3. Results Framework

3.1. Targets for Joint SDG Fund Results Framework

Joint SDG Fund Outcome 2: Additional financing leveraged to accelerate SDG achievement

Private Finance Leverage	Private Funds	SDG Fund	Public Funds	Private Funds Leverage	Public Funds Leverage
Outcome 1	$ 31,400,000	$ 3,910,000	$ 800,000	**8.0 : 1**	0.2 : 1
Outcome 2	$ 6,500,000	$ 2,600,000	$ 0	**2.5 : 1**	0.0 : 1
Total	$ 37,900,000	$ 7,800,000	$ 800,000	**4.9 : 1**	0.1 : 1

Joint SDG Fund Operational Performance Indicators

- (do not change or add – this is for information only so that teams know what they will be assessed against) Level of coherence of UN in implementing programme country[39]
- Reduced transaction costs for the participating UN agencies in interaction with national/regional and local authorities and/or public entities compared to other joint programmes in the country in question
- Annual % of financial delivery
- Joint programme operationally closed within original end date
- Joint programme financially closed 18 months after their operational closure
- Joint programme facilitated engagement with diverse stakeholders (e.g., parliamentarians, civil society, IFIs, bilateral/multilateral actor, private sector)
- Joint programme included addressing inequalities (QCPR) and the principle of "Leaving No One Behind"
- Joint programme featured gender results at the outcome level
- Joint programme undertook or draw upon relevant human rights analysis, and have developed or implemented a strategy to address human rights issues
- Joint programme planned for and can demonstrate positive results/effects for youth
- Joint programme considered the needs of persons with disabilities
- Joint programme made use of risk analysis in programme planning
- Joint programme conducted do-no-harm / due diligence and were designed to take into consideration opportunities in the areas of the environment and climate change

[39] *Annual survey will provide qualitative information towards this indicator.*

3.2. Joint Programme Results framework

Result / Indicators	Baseline	2022 Target	2023 Target	2024 Target	2025 Target	Means of Verification	Responsible partner
Outcome 1							
Outcome Indicator 1.1 Expected annual energy saved: 81,000 MWh	0	8,100	36,450	32,400	4,050	Quarterly reports from the project and verification consultants	EBRD
Outcome Indicator 1.2 Estimated GHG emissions avoided (tCO2e/year): 75,654 tonnes	0	7,565	34,044	30,262	3,783	Quarterly reports from the project and verification consultants	EBRD
Outcome Indicator 1.3 Renewable capacity added: 8.1 MW	0	0.81	3.65	3.24	0.41	Quarterly reports from the project and verification consultants	EBRD
Output 1.1							
Output 1.1.1 Number of sub-loans extended by banks: 105 (sex-disaggregated)	0	10	50	40	5	Quarterly reports from banks	EBRD
Output 1.1.2 Total value of sub-loans extended: $31.4M	0	3,000,000	15,000,000	12,000,000	1,400,000	Quarterly reports from banks	EBRD
Output 1.2							
Output 1.2.1 Total value of performance-based payments allocated: USD 3.14 M	0	300,000	1,500,000	1,200,000	140,000	Reports from Project Consultants, Verification Consultants and banks	EBRD
Output 1.3							
Output 1.3.1 Number of loan officers trained to assess RE/EE project loan applications: at least 20 (sex-disaggregated)	0	17	3			Reports from Project Consultants, and banks	EBRD
Output 1.3.2 Number of loan officers trained on gender-disaggregated data	0	17	3			Reports from Project Consultants, and banks	EBRD

collection and analysis: at least 20							
Output 1.3.3 Number of businesses reached through marketing efforts: at least 700 (sex-disaggregated)	0	200	300	150	50	Reports from Project Consultants, and banks	EBRD
Outcome 2							
Outcome Indicator 2.1 Expected annual energy saved: 5,200 MWh	0	520	2,340	2,080	260	Quarterly reports from the project and verification consultants	UNDP & IOM
Outcome Indicator 2.2 Estimated GHG emissions avoided (tCO2e/year): 4,856.8 tonnes	0	486	2,186	1,943	243	Quarterly reports from the project and verification consultants	UNDP & IOM
Outcome Indicator 2.3 Renewable capacity added: 2.6 MW	0	0.26	1.17	1.04	0.13	Quarterly reports from the project and verification consultants	UNDP & IOM
Output 2.1							
Output Indicator 2.1.1 Total value of loans extended: $6.5 M	0	400,000	3,000,000	2,700,000	400,000	Quarterly reports from banks	UNDP & IOM
Output Indicator 2.1.2 Number of loans extended by banks: 650 (sex-disaggregated)	0	40	300	270	40	Quarterly reports from banks	UNDP &IOM
Output Indicator 2.1.3 Number of loans extended to women and women-headed households (including returning migrants and remit: 227	0	10	100	100	17	Quarterly reports from banks	UNDP
Output Indicator 2.1.4 Number of loans extended to returning migrants and remittance recipients: 150 (sex-disaggregated)	0	5	70	65	10	Quarterly reports from banks	IOM
Output 2.2							
Output Indicator 2.2.1 Total value of all performance-based payments allocated: $1.95 M	0	120,000	900,000	810,000	120,000	Quarterly reports from banks	UNDP&IOM

Output Indicator 2.2.2 Total value of performance-based payments allocated to returning migrants and remittance recipients: $ 0.45 M	0	15,000	210,000	195,000	30,000	Quarterly reports from banks	IOM
Output 2.3							
Output Indicator 2.3.1 Number of loan officers trained to assess RE/EE project loan applications: at least 20 (sex-disaggregated)	0	17	3			Quarterly reports from banks	UNDP & IOM
Output Indicator 2.3.2 Number of loan officers trained on gender-disaggregated data collection and analysis: at least 20	0	17	3			Quarterly reports from banks	UNDP & IOM
Output Indicator 2.3.3 Number of individuals reached through marketing efforts: 1000 (sex-disaggregated)	0	286	429	214	71	Quarterly reports from banks	UNDP & IOM

Annex 4. Gender marker matrix

Indicator N°	Indicator Formulation	Score	Findings and Explanation	Evidence or Means of Verification
1.1	Context analysis integrate gender analysis incorporating use of sex disaggregated data	3	CO performed gender analysis across of the financial sector and identified underlying causes of gender inequality and discrimination in line with SDG priorities including SDG 5. Furthermore, sex-disaggregated and gender sensitive data is included wherever available and the CO conducted research of the women 65+ years and the challenges of women headed households in converting to renewable energy sources, as a gender analysis of those furthest behind. Sex disaggregated data is integrated in all baselines. The background analysis includes the underlying causes of gender inequality.	Internal gender background analysis report, research on the financial sector and researched done in gender in climate change.
1.2	Gender Equality mainstreamed in proposed outputs	2	One programme output specifically targets gender equality in line with SDG priorities including SDG 5	SDG application
1.3	Programme output indicators measure changes on gender equality	1	9 indicators, or 32%, of the output indicators measure changes in gender equality and the empowerment of women in line with SDG targets including SDG 5.	SDG application
2.1	PUNO collaborate and engage with Government on gender equality and the empowerment of women	3	As a continuation of the close collaboration with Ministry of Labor and Social Policy and Ministry of Environment and Spatial Policy, there were consultative meetings held with the state secretaries from both ministries in the process of project preparation. Furthermore, the National Women's Machinery was consulted in country analysis, strategic prioritization, and planning of the implementation and M&E.	PUNO internal documents, minutes from meetings
2.2	PUNO collaborate and engages with women's/gender equality CSOs	3	There is continuous collaboration with women CSOs on diverse topics and in the recent several years there is more intense collaboration on the topic of gender equality and climate change.	PUNO internal documents, minutes from meetings
3.1	Programme proposes a gender-responsive budget	2	31% of the total budget is allocated to gender equality or women's empowerment.	SDG application
Total scoring		**2.3**		

Annex 5. Risk Management Plan

The risk management strategy that will be deployed in this project takes a systematic and structured approach. The experience of EBRD in developing and executing similar projects has provided a wealth of information about the possible risks that could be faced and appropriate mitigation measures. Additionally, UNDP, together with IOM, have a rich experience in designing and executing projects in the country which have informed project design. The approach to risk management has been to design the project in a manner that minimizes risks and yet leaves space for an iterative and adaptive process, where risks and mitigation measures are reconsidered as new information becomes available. At each step along the program, the project management and oversight teams will build upon the results of the assessments done at previous stages. New risks may become apparent as the project progresses and areas of programmatic focus or approach become more detailed. At the same time, the approach to risk management is detailed enough to inform decision-making but not overly burdensome. The risk management table below identifies the expected possible risks and mitigation measures.

Risks	**Risk Level:** Likelihood x Impact)	**Likelihood:** Certain - 5 Likely - 4 Possible - 3 Unlikely - 2 Rare – 1	**Impact:** Essential – 5 Major - 4 Moderate - 3 Minor - 2 Insignificant - 1	Mitigating measures	Responsible Org./Person
Contextual risks					
Limited capacity of the market to provide high-quality products and services to meet the expected increased demand leading to insufficient supply and delays in implementation.	8	2	4	Project consultants will conduct consultations with local providers of RE & EE solutions early in the implementation of the project to inform them of the project and expected increased demand for products and services, encouraging them to prepare.	Project consultants with guidance and oversight from UNDP, EBRD & IOM
Limited uptake of the product by target beneficiaries	10	2	5	Awareness campaigns will be reassessed and increased to make sure that target beneficiaries are aware of the project. The awareness campaign will customize the approach towards project target groups with full understanding of their challenges and in close cooperation with stakeholder groups, such as economic chambers, CSOs, financial institutions and government bodies.	Project consultants and participating banks with guidance and oversight from UNDP, IOM & EBRD

Remittance recipients and returning migrants are unable or unwilling to participate in the project.	12	3	4	Public awareness raising by IOM, marketing campaign by local banks and work of consultants as part of TA should provide additional information that will help beneficiaries better understand their return on investment, benefits of RE and EE solutions and positive impact in terms of quality of life including air quality. Also, early adopters' success, promotion of successful case studies and subsequently peer learning may also play a role in wider uptake of the product.	IOM
Covid-19 results in a sustained economic recession reducing companies' and households' ability to invest in RE and EE solutions.	12	3	4	EBRD's experience shows that the demand for grant supported financing did not decrease due to Covid. On contrary, the participating banks are signaling stable demand, which led to signing four new agreements for both the retail and corporate clients, despite the Covid. The project will continue to monitor situation closely and take early actions and adjustments as needed.	UNDP, IOM & EBRD
Programmatic risks					
Limited collaboration and coordination between stakeholders slow project implementation and limits effectiveness.	8	2	4	Conduct extensive stakeholder consultations during project development to secure buy-in from project partners and ensure coordination of project interventions. The project manager will carefully engage in and closely monitor coordination during implementation.	UNDP & RCO
Local banks delay their participation in the project.	10	2	5	Interviews have been conducted with the local banks and they showed strong interest to participate in GFF, once established. The demand for EBRD's similar products has remained stable despite the crisis. Situation will be closely monitored and mitigation actions taken.	UNDP & EBRD
Institutional risks					

Institutional capacity/coordination of executive entities results in inadequate human resources to effectively manage project implementation.	6	2	3	Project Management team to conduct a stock take of capacities of the various entities involved in delivering the programme to identify issues early on and take actions to mitigate any issues.	UNDP & IOM
Fiduciary risks					
Scale of loan borrowers default on repayments.	9	3	3	The imposed repayment moratoria due to the Covid crisis disguises the real situation with the NPLs in the Macedonian banking sector. The real effects will be seen at end 1Q 2021, when the 2nd repayment moratoria will end. This is mitigated by the fact that the largest banks on the market are reporting much better results than the initial forecasts at the start of the Covid crisis and this is mostly based on the expectations that the NPLs will remain low. The 2nd repayment moratoria targeted only clients that have ongoing effects from the Covid crisis, and their number is minimal compared to the clients that used the first moratoria.	Individual Participating Banks with close oversight and support from EBRD

Annex 6. Monitoring and Evaluation Plan

The project will be monitored through the following monitoring and evaluation plan:

Monitoring Activity	Purpose	Frequency	Expected Action
Track results progress	Progress data against the results indicators in the RRF will be collected and analyzed to assess the progress of the project in achieving the agreed outputs.	Annually	Slower than expected progress will be addressed by project management.
Monitor and Manage Risk	Identify specific risks that may threaten achievement of intended results. Identify and monitor risk management actions using a risk log. This includes monitoring measures and plans that may have been required as per Social and Environmental Standards. Audits will be conducted in accordance with UNDP's audit policy to manage financial risk.	Quarterly	Risks are identified by project management and actions are taken to manage risk. The Risk Log is actively maintained to keep track of identified risks and actions taken.
Learn	Knowledge, good practices and lessons will be captured regularly, as well as actively sourced from other projects and partners and integrated back into the project.	At least annually	Relevant lessons are captured by the project team and used to inform management decisions.
Annual Project Quality Assurance	The quality of the project will be assessed against quality standards to identify project strengths and weaknesses and to inform management decision making to improve the project.	Annually	Areas of strength and weakness will be reviewed by project management and used to inform decisions to improve project performance.
Review and Make Course Corrections	Internal review of data and evidence from all monitoring actions to inform decision making.	At least annually	Performance data, risks, lessons and quality will be discussed by the project board and used to make course corrections.

Project Report	A progress result-oriented and evidence-based report will be presented to the Project Board. It will consist of progress data showing the results achieved against pre-defined annual targets at the output level, the annual project quality rating summary, an updated risk long with mitigation measures, and any evaluation or review reports prepared over the period. UNDP will submit any other reports as required to the SDG Fund in line with the reporting requirements set out in the Joint SDG Fund Operational Guidance.	Annually	
Project Review (Project Board)	The project's governance mechanism (i.e., Project Steering Committee) will hold regular project reviews to assess the performance of the project and review the Multi-Year Work Plan to ensure realistic budgeting over the life of the project. In the project's final year, the Project Board shall hold an end-of project review to capture lessons learned and discuss opportunities for scaling up and to socialize project results and lessons learned with relevant audiences.	At least annually	Any quality concerns or slower than expected progress should be discussed by the project board and management actions agreed to address the issues identified.

Annex 7. Communication Plan

1) Overall narrative of the joint programme

The programme seeks to increase SME and household energy efficiency (EE) and use of renewable energy (RE) in North Macedonia through financial incentive schemes designed to catalyze investments into EE and RE solutions. Through the creation of a Green Financing Facility (GFF), the programme seeks to accelerate investments into RE and EE by providing a flexible and lean financial mechanism that connects multiple interventions to unlock private capital flows from financial institutions, SMEs and households for the acquisition of RE and EE technologies, resulting in increased SME competitiveness, reduced greenhouse gas (GHG) emissions and air pollution, increased efficiency in energy consumption, as well as in enhanced local market innovation and linkages.

2) Strategic approach to key audiences

Advocacy

- **Key audience: Local organizations that work with targeted groups** (female-headed households, single-parents, households with PWD, Roma, remittance recipients, returning migrants and employees affected by COVID-19)
    - Engage local organization to help promote the GFF services to the target groups they serve by presenting a case for how the GFF will benefit their beneficiaries and the portfolios of their organizations.
    - Support local organizations in reaching out to the target groups by funding tailor-made promotional content, including those focused on behavioral approaches, and activities that will assist them in the task.
    - Incentivize local organizations to participate in outreach and promotion by crediting them for their contribution to the wider programme communication efforts.
    - Incentivize local organizations to participate presenting this as an opportunity for them to build their capacities ahead of the Green Agenda for Western Balkans, which will be launched under the European Green Deal.

Media and promotion

Note: Media, marketing and promotion activities linked to the Outcome 1 and 2 financial products will be financed by the local banks offering them. The implementing organizations will support their efforts as needed, particularly in reaching specific target groups.

- **Key audience: Individuals and SMEs**
    - Stimulate interest in RE and EE investment through the GFF
        - Use all available media channels to promote the economic and environmental benefits of RE and EE investment through the GFF, using real testimonies of early adopters whenever possible in combination with projected (and later actual) ROI data.
        - Create opportunities for GFF clients (especially first-movers) to promote them RE and EE investments as not only business savvy (as they are increasing SME's competitiveness), financially prudent but also virtuous decisions. Support SMEs in generating positive publicity for their businesses by promoting their green investments.

        - Create a positive public narrative once sufficient critical mass of GFF clients is available. Encourage a public narrative that RE and EE investments are a rising trend that business that want to compete at the European market as well as responsible businesses and households need to become a part of if they care about the environment and their success on the market.
        - Ensure that female heads of households and owners of businesses are adequately represented in all communications materials and activities.
        - Strengthen messaging to female heads of households and owners of businesses by co-designing them with the help of relevant women's organizations.
- **Key audience: Target groups** (female-headed households, single-parents, households with PWD, Roma, remittance recipients, returning migrants and employees affected by COVID-19. Within the groups of remittance recipients and returning migrants female headed households, single parents, households with PWD, employees affected by COVID–19 will be prioritized)
    - Stimulate interest in RE and EE investment through the GFF
        - Use all available media channels to promote the economic and environmental benefits of RE and EE investment through the GFF, using real testimonies from representatives of each target group whenever possible in combination with projected (and later actual) ROI data. For groups/communities with limited media infrastructure penetration, focus promotion efforts on channels that have proven effective in reaching such groups in the past.
        - Engage "champions" (e.g. eminent/popular personalities) for different target groups to raise awareness about the RE and EE solutions.
        - Develop informational, educational and promotional contents that address the specific concerns or knowledge gaps of each target group that prevent them from increasing their investment in RE and EE.
        - Reach out to target groups through trusted local organizations with a solid track record of providing value for said groups.
        - Develop the informational, educational and promotional contents to stimulate investment in RE/EE for the returning migrants and remittance recipients and prioritized groups within these two groups, based on behavioral insights on air pollution, remittance use and adoption of clean energy technologies; recognition of the trans-local nature of household decision-making process (including gender roles), peer networks, specific channels of wire transfers or receipt of money and use of social media.

Public relations

- **Key audience: General public in North Macedonia**
    - Shape the narrative around the GFF as a transformational activity that is strategically positioned to accelerate adoption of energy efficiency and renewable energy solutions and their impact on SME competitiveness, GHG

emissions and air pollution as one of the undisputedly top issues facing the country.
    - Communicate how the intersectoral partnership and multi-agency model of the GFF will help to ensure replicability/(sub)regional scale up, success and sustainability of the intervention.
    - Encourage the public to see itself not as a passive observer, but as an agent of change in the realm of EE, RE, air pollution and greenhouse gas emission reduction. Channel willingness to affect change into action through the opportunities made available by the GFF.
- **Key audience: Regional and global public**
    - Highlight success milestones of the GFF as evidence of how this approach to increasing EE and RE is effective at achieving significant progress on complex national and global issues by laying the foundation for sustainable systems that can lead to transformational change.
    - Highlight the "geographically agnostic" elements at the core of the GFF as evidence that it can be successfully scaled and replicated elsewhere.
    - Highlight the importance of addressing economic barriers to achieving environmental objectives, from the individual to the national and global level.
    - Highlight that the GFF will contribute to the Green COVID – 19 recovery and Green Agenda for Western Balkans, which will be launched under the European Green Deal.
- **Key audience: Investors**
    - Highlight the importance of the Joint SDG Fund in catalyzing transformative solutions to complex national and global issues related to achieving the SDGs through case studies and success stories that demonstrate how funding translates into real impact.
    - Highlight the ease of replication and scalability of the project's model by plotting its growth trajectory in the country and focusing on the "geographically agnostic" elements at the core of the GFF.
    - Communicate the importance of additional investments after the project's lifecycle for keeping up the momentum of change and ensuring sustainability.
    - Support each organization in advancing communications goals adjacent to those of the joint programme by generating suitable content as part of the programme that they can use in their wider communications activities.
- **Key audience: Partners**
    - Communicate how the intersectoral partnership and multi-agency model of the GFF are essential to its replicability/(sub)regional scale up, sustainability and success.
    - Communicate good project management and implementation practices.
    - Highlight the importance of working together to reach a diverse range of target groups and ensure that no one is left behind.
    - Support each organization in advancing communications goals adjacent to those of the joint programme by generating suitable content as part of the programme that they can use in their wider communications activities.

3) Objectives of strategic communication plan

Objective	Indicator (gender-disaggregated where relevant)	Measurement method
Maximize the number of local organizations supporting GFF outreach to targeted vulnerable groups	Number of local organizations supporting GFF outreach to targeted groups	Internal records
Maximize the number of SMEs reached with information the GFF financial products for GE/EE investments.	At least 500 businesses reached through marketing efforts	Survey
Maximize the number of individuals reached with the GFF financial products for GE/EE investments (differentiated by the target groups)	At least 750 individuals reached through marketing efforts	Survey
Increase local media coverage of RE and EE investments and innovations as key solutions to local environmental issues and family well-being	Percentage increase in the number of media products discussing RE and EE at the household and SME level	Baseline desk review of news stories and other media content on RE and EE investments followed up by a desk review the end of the programme (supported by media clipping/analysis company)
Generate informational, educational and promotional content based on programme results that investors and partners can use to advance their communications objectives	Number of communications activities and products delivered to partners and investors for the needs of their strategic communication objectives	Internal records

4) Main activities

Activity	Channel	Focal point	Resources	Milestones
Meetings, presentations and other events focused on getting local organizations to support GFF outreach to targeted groups (special focus on organizations that work with female-headed households, migrants and SMEs)	Online conferencing tools or physical meetings/events	UNDP Programme Communications Specialist	500 USD	50% of targeted organizations reached 100% of targeted organizations reached
Production of informational,	Social media, online advertising,	UNDP/ IOM	8,000 USD	One major communications

educational and promotional content on GFF funded by the implementing organizations, tailored to different channels and target audiences	mailing lists, online/offline presentations, events, print materials (limited), online videos, media articles	Programme Communications Specialist		product developed per target audience Three major communications products developed per target audience
Production of informational, educational and promotional content on GFF funded by partner financial institutions, tailored to different channels and target audiences	Television, radio, outdoor advertising, social media, online advertising, mailing lists, online/offline presentations, events, print materials (limited), online videos, media articles	UNDP/ IOM Programme Communications Specialist	Resources mobilized from local financial institutions	Three major communications products developed per target audience Six major communications products developed per target audience
Monitoring activities	Internal record keeping, surveys, desk reviews	UNDP Programme Communications Specialist	12,000 USD	All monitoring reports finalized

Annex 8. Learning and Sharing Plan

The Learning and Sharing plan will rely three pillars:

1. Continuous data gathering and analytics.
2. Online platform for easy access and sharing of data, information and knowledge between partners.
3. Regular "Pause and Reflect" meetings between partners to share lessons learned from progress on implementation, discuss challenges, propose solutions and to take the time to "zoom out" and consider the big picture of the programme.

In terms of substance and content, the Learning and Sharing plan will focus providing actionable insights on the GFF as a model, strategic partnerships, and project results and development outcomes.

1) Strategic approach to learning and sharing

1. Performance data on programme indicators will be continuously collected through a variety of tools (primarily surveys, internal records and official statistics), integrated and analyzed to provide insights relevant to all partners. This will include project relevant data sourced from the partner banks.
    a. Lessons learned will be identified through a combination of quantitative data gathering and analysis, qualitative interviews and informal conversations with key stakeholders, as well as discussions during "Pause and Reflect" meetings between partners. They will be captured in research reports, analytical blog posts and interview recordings/ podcasts.
    b. Learning resources will be systematized and consolidated on online platforms (primarily SharePoint, Teams and Trello) that will be accessible to all partners.

c. The platform will also be used to share learning resources and facilitate learning, together with regular "Pause and Reflect" meetings.
d. Key findings will be shared with programme and policy decisionmakers in tailor-made products with explicitly stated implications and recommendations specific to organizations/institutions.

2) Objectives of learning and sharing

Objective	Indicator	Measurement Method
Ensure that all partners have access to the same data and learning resources	Number of partners with access to shared data and learning resources	Internal records
Active partner participation in learning activities	Number of partners participating in learning activities (e.g. research, analysis, blogging, podcasts, "Pause and Reflect" meetings, etc.)	Internal records
Partners and other key stakeholders using learning resources to inform their strategic decisions	Number of partners and key stakeholders using learning resources to inform their strategic decisions	Survey
Learning resources remain available to all interested stakeholders after the program's conclusion	Accessibility of learning resources [qualitative]	Accessibility assessment of knowledge management platform and interviews with stakeholders on ease of access

3) Main activities

Activity	Tools	Focal point	Resources	Milestones
Set up an online learning platform	SharePoint, Teams and Trello	UNDP Monitoring Specialist	n/a (listed tools are standard in UN agencies or are free to use)	Online platform is fully operational
Draft learning and sharing (L&E) plan	Standard office software	UNDP Monitoring Specialist	n/a (task of monitoring specialist)	L&E plan finalized
Hold "Pause and Reflect" meetings	Programme calendar	UNDP Monitoring Specialist	n/a (task of monitoring specialist)	50% of planned meetings held 100% of planned meetings held
Conduct data collection, analysis and produce customized learning resources for different target segments	Surveys, interviews, data analysis, written reports, presentations, blog posts, podcasts, videos, case studies, success stories, fact sheets	UNDP Monitoring Specialist	USD 10,000	50% of planned learning resources finalized 100% of planned learning resources finalized

Annex 9. Safeguards and ESG standards

In terms of safeguards and ESG standards for **Participating Financial Institutions (PFIs) under outcome 1 and outcome 2**, **all PFIs must comply with the** EBRD's 2019 Environmental and Social Policy (ESP)**;** conduct its business in accordance with EBRD Performance Requirements 2 on Labor and working conditions and 4 on Health, Safety and Security; have in place an environmental and social management system in accordance with EBRD Performance Requirement 9 on financial intermediaries, including an environmental and social policy and procedures commensurate with the nature of the Borrower, and the level of environmental and social risks associated with the Sub-Projects. In addition, all on-lending activities must be in compliance with the relevant national environmental, health and safety, social and labor standards and regulations.

EBRD-financed projects are expected to be designed and operated in compliance with good international practices relating to sustainable development. To help EBRD clients and their projects achieve this, EBRD has defined performance requirements covering the key areas of environmental and social issues and impacts those projects are required to meet. The performance requirements provide a solid base from which clients can improve the sustainability of their business operations.

Annex 10. Market study – aim, scope and methodology

In order to further substantiate project proposal and test various project assumptions, a market study was contracted and independent consultants developed analysis that was used to respond to the feedback of independent evaluators to the Joint Programme that was submitted to the SDG Fund.

Aim and scope of the study:
The Market Study assessed the underlying enabling environment and broad impact factors that led to the conceptualization of the GFF, shedding light to key questions and clarification request raised by the Joint SDG Fund. The Study was carried out by the Frankfurt School of Finance & Management GmbH in cooperation with the Macedonian Center for Energy Efficiency and is based on the latest available information collected directly in the country by local experts during the 1 July – 20 September 2021 reporting period.

The Study presents the market research results together with the expert analysis and includes:

- Assessment of the adequateness of the proposed pricing level of PBPs for GFF;
- Evaluation/comparison of the effectiveness of PBP vis-a-vis other models of financial risk sharing in increasing access to credit finance for RE/EE via banks;
- Assessment of the relevance of inclusion of project and verification consultants as an integral part of the GFF and their impact on effectiveness and efficiency of the GFF;
- Assessment of the plausibility of inclusion of ESCOs/EaaS as part of the GFF implementation structure; and
- Recommendations on how UN and EBRD can further enhance the structure of GFF.

Implementation approach and methodology:
The methodology of the Study is based on desk research, with the collection of primary sources through direct, semi-structured interviews, and data analysis. Primary sources include, but are not limited to, documents pertaining to activities of relevant governmental, non-governmental and international organizations regarding their priorities in the private sector and households. The data collection efforts included the following.

Implementation Approach and Methodology
Approach: Deductive/Inductive logic based on quantitative and qualitative data. The methodological approach followed is composed of the three main steps: 1) Identification of data sources; 2) Data collection; and 3) Data analysis and recommendations.
Identified data sources: 1. Governmental documents (state strategies, regulations, intl. treaties and agreements) 2. Relevant studies, articles, research papers 3. Reports or documents containing statistical data on the operation or economy of some sectors (e.g. from State Statistical Office, National Bank of Republic of North Macedonia, IEA, Chamber of Commerce, etc.) 4. Relevant sector-practitioners and real-sector representatives willing and able to share relevant information. Two representative sample surveys were conducted, in particular MSME research with 200 respondents and household research with 450 respondents. In depth interviews were performed with 15 MSMEs and 20 target households. In addition, a focus group with 11 energy auditors and verification consultants was conducted. **Data collection:** 1. Desktop-based research and collection of relevant documentation and data

2. Personal interviews with relevant stakeholder such as ministries and related Agencies
3. Personal interviews with finance-related experts / specialists and representatives of local commercial banks and public finance institutions (MBDP, FIDT, BAU)
4. Experts' opinion
5. Primary research for target households
6. Primary research for target private companies

Data analysis and recommendations:

1. Summary of key data according to thematic areas (regulatory, technical, and financial)
2. Conclusion and recommendation
3. References and data sources
4. Supporting information (Annexes)

The approach, based on a triage of data research methodology included primary data collection through direct interviews, secondary data collection through desk research, and experts' inputs and professional opinions provided for relevant data on the PBPs. Lastly, the methodology is based on successful regional and international examples in carrying out similar studies related to renewable energy and energy efficiency financing initiatives and case studies of implemented RES/EE projects.

For the purposes of primary data collection, three questionnaires were developed. In particular, semi structured questionnaire for the financial institution that was emailed following the individual meetings. Total of four responses were collected during the month of July 2021, from Komercijalna Bank AD Skopje, NLB Bank AD Skopje, Stopanska Bank AD Bitola, and Halkbank AD Skopje. Findings were enriched with the information gathered over the meetings with the other banks, such as Silk Road Bank AD Skopje, Pro Credit Bank AD Skopje, Sparkasse Bank AD Skopje, Capital Bank AD Skopje as well as Development Bank of North Macedonia JSC Skopje.

Furthermore, the private sectors' opinion on EE/RE investments and experience with TA/VC was explored via dedicated research instrument and stratified random sampling. The MSMEs survey was designed including closed and opened ended questions, aimed at collecting data that would provide information for both quantitative and qualitative analysis accordingly. Demographic data of the company are included, sections to explore the already implemented EE/RE measures, if any, the intentions for future investment in EE/RE, section on sources of financing and funding awareness, and section for TA/VC and grants experience. Where possible, the survey was administered via face to face (F2F) interviews otherwise via a scheduled telephone call, during the months of July and August 2021. The sample has combined 16 sub-industrial sectors[40] into a population size estimated at 1,900 companies whereas the representative sample targeted 200 companies with marginal error of 5% and confidence interval od 95%. The sectors were selected based on how energy intensive their processes are, past record in the implementation of the EE/RE investments, and expert opinion on where there is the highest energy saving potential. The response rate was 60% mainly due to the holiday season, lCOVID-19 restrictions and the technical nature of the survey. Additionally, 15 in-depth interviews were performed with companies that have already invested in EE/RE on the basis of the same research instrument but focusing more on the open-ended questions.

Designed to gather data for the Component 2, the primary data collection was also performed among the target households via questionnaire using stratified random sampling. The instrument included demographics about the respondents and the household, sections aiming at exploring past and potential EE/RE investment, decision making and attitude towards EE/RE, awareness of sources of finding and funding awareness, section for TA/grants

[40] Source: Central Registry of North Macedonia, 2021

experience as well as dominant channels of information. Where possible, the survey was administered via F2F interviews otherwise via a scheduled telephone call, during the month of July and August 2021. Few target groups were pre - selected and were contacted with assisted help from associations of citizens, non-formal groups of people and partner organizations.

In the country, there are about 500,000 households in total, of which 10% were estimated to belong to the population targeted by the project. or 50,000. In particular, households that belong to the category mother with children are estimated at 31074, households of father with children 7,914, employed disabled persons (4,000) in private companies for employment of disabled persons as well as 2,000 -3,000 returning migrants per year based on the IOM administration. The representative sample of 380 household was calculated with 5% error of margin and 95% of confidence interval. Yet, the sample was increased to 450 households to gather even more relevant information. The response rate was 60%, so practically over 650 households were contacted. Additionally, 20 in-depth interviews were performed with household-representatives of each of the target groups on the basis of the same research instrument by dominantly focusing on the open-ended questions.

To further explore the process for acceptance and behavior towards PBPs, a focus group, comprised of 10 energy auditors and verification consultants that have FF expert experience in TA or project verification programs, was conducted in August 2021.

Additional information were collected through cross-sectoral interviews with the relevant institutions: Ministry of Economy (Energy Sector and Sector for Industrial and Investment Policy), the Ministry of environment and physical planning, the Fund for Innovation and Technological Development, the City of Skopje (Unit for Energy, EE and RE/Department for Environmental Protection), the Ministry of Finance (including representatives from CFCD), the REEP/EU project Creating conditions for the implementation of ESCO projects, the EBRD/EU SME Competitiveness Programme and the EBRD GEFF Project. The research team also relied on expert knowledge of team members as few of them have extensive experience in working on implementation of similar financing facilities in both North Macedonia and the region. In fact, the team members have directly participated in the implementation WeBSEFF I and II, GGF and SME Competitiveness Project and have contributed their insights.

Annex 11. Research on the behavioral barriers and enablers of household spending on energy-efficient and renewable energy solutions in North Macedonia – Scope and methodology

Scope of the study:

A behavioral study was conducted to gain better/deeper understanding of barriers and enablers of household spending on EE and RR solutions among all target groups (i.e. female-headed households, single-parent households, households with a person with one or more disabilities, Roma households, households with at least one member whose job has suffered due to the COVID-19 pandemic, remittance recipient households and households with returning migrants) with the aim to inform the design of the Green Financing Facility (GFF) – Outcome 2. Namely, as GFF is targeting above mentioned marketable but underserved groups it was beneficial for Facility design to understand if and under which conditions they would be ready to take the subsidized loans to fund the purchase and adoption of EE and RE solutions.

A scoping study was undertaken. It drew from the disciplines of environmental and health psychology and sociology, and a theoretical model from applied social psychology called the Model of Private Proactive Adaptation to Climate Change (MPPACC). It was developed by Grothmann and Patt (2005), expanded by Dang, Li and Brewer (2012)[41] and additional variables added by Valkengoed and Steg (2012)[42] to explain the factors that motivate individuals and households to engage in the desired behavior that successful project implementation requires.

Methodology:

The study was comprised of: quantitative and qualitative components. The quantitative component took the form of a face-to-face household questionnaire survey (sample size: 1188 respondents) for which Valkengoed and Steg's (2019) set of 15 motivational factors (*behavioural predictors*) for adapting to climate change was adapted for conceptual and technical reasons. Each one of these motivational factors was adapted to take the form of a psychometric proxy indicator employing a multi-item scale of statements to which respondents had to indicate their response, such as whether or not they agreed with the statement made, i.e. on a 5-point Likert scale. The survey also collected demographic data from all respondents. The dataset was then analyzed using standard statistical tests used in quantitative social psychology research of: 1) internal consistency/reliability (Cronbach's Alpha), basic construct validity (Pearsons R) and correlation and regression analyses (Multivariate Linear Regression) by the whole survey sample, and then by each of the seven target groups.

The qualitative component consisted of seven pre-scripted focus group discussions (FGDs), one per each of the seven population groups of interest, conducted over the remote online communications platform (*Zoom*) due to the ongoing COVID-19 pandemic. Each FGD was moderated by an experienced qualitative researcher, and employed the same script comprised of semi-structured questions with moderator prompts to tap into issues of interest.

[41] Grothmann, T., & Patt, A. (2005). Adaptive capacity and human cognition: The process of individual adaptation to climate change. Global Environmental Change, 15, 199–213. http://doi.org/10.1016/j.gloenvcha.2005.01.002 , Dang, H. L., Li, E., Nuberg, I., & Bruwer, J. (2014). Understanding farmers' adaptation intention to climate change: A structural equation modelling study in the Mekong Delta, Vietnam. Environmental Science & Policy, 41, 11–22. http://doi.org/10.1016/j.envsci.2014.04.002

[42] Anne van Valkengoed and Linda Steg (2019), The Psychology of Climate Change Adaptation. Elements in Applied Social Psychology (ed. Susan Clayton). Cambridge University Press.

The question schedule was based on an adapted form of the questions used in the quantitative survey to provide the potential for limited triangulation of quantitative with qualitative results, and/or cross-verification of trends, exceptions, anomalies and other nuances in both types of data. An additional FGD was held with the 'early adopters' group, and 4 key informant interviews (KIIs) were held with representatives of the Ministry of Economy, UNDP and commercial banks to obtain their institutional perspectives on the scheme and its potential outcomes

Annex 12. Theory of Change Diagram

Goal
North Macedonia advances towards a low-emission sustainable development pathway via the utilization of renewable energy and energy-efficient solutions, benefiting both women and men and target groups.
Goal Statement
If access to finance and incentives are improved, together they will trigger investments into renewable energy and energy efficiency solutions that will contribute to improved air quality, reduced CO2 emissions, and sustained socio-economic growth.
Outcomes
1. SMEs have greater access to finance for accelerating a transition to renewable energy and energy efficiency solutions. 2. Target groups/households, especially women and women-headed households, have greater access to finance for accelerating a transition to renewable energy and energy efficiency solutions.
Barriers
1. Local financial institutions have limited interest in lending for the acquisition of RE and EE solutions. 2. Limited willingness of SMEs and individuals to prioritize investments in RE/EE. 3. Low absorption of financial instruments by women and women-owned companies. 4. Target groups perceive RE as costly and complicated. 5. Remittance recipients have adopted EE solutions but are not prioritizing investments in RE solutions.
Assumptions
• Local Banks, suppliers, and service providers are willing to participate in the project. • SMEs and households are financially able & triggered to invest in RE and or EE solutions given the new financial instruments that include incentives and technical support mechanisms. • Targeted stakeholders are made aware of the project.